



Optimized deep learning model for comprehensive medical image analysis across multiple modalities

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ABSTRACT

This study presents a novel amalgamated model for the diagnosis of multiple medical conditions using various imaging modalities, including Chest X-ray, MRI, and endoscopic images. Each imaging modality has unique protocols and feature characteristics, presenting significant complexity and challenges in diagnosis. To address these issues, we propose a truncated lightweight model that effectively fuses diverse image features while reducing computational requirements. The model utilizes deep learning (DL) techniques and multi-scale feature learning to enhance diagnostic capabilities across different medical images. Specifically, it employs an efficient MobileNet architecture to simultaneously diagnose multiple diseases. Key innovations include model truncation, a modified naive inception block for multi-scale feature extraction, and metaheuristic optimization methods. By optimizing the architecture with techniques such as Chain Foraging and Cyclone Aging, the model achieves robustness and scalability, improving generalization across varying image resolutions. Additionally, an integrated ConvLSTM unit before the Softmax layer enhances feature extraction across spatial and temporal dimensions, addressing challenges related to differing feature sizes and scales in multi-disease diagnosis. We conducted comprehensive testing on publicly available multi-class medical image datasets, including brain MRI, chest X-ray, and gastro endoscopic images, which demonstrates that the proposed model outperforms existing methods, achieving an overall accuracy of 97.37 %. To further support clinical decision-making, we utilized visualization techniques such as GradCAM, and Feature Map analysis to enhance the interpretability of the model predictions. Overall, the proposed model not only showcases exceptional performance in classifying various medical images but also presents an optimal balance between accuracy and computational efficiency, establishing its potential as a practical solution for accurate multi-modal medical image analysis.

1. Introduction

Medical image classification is a key factor in healthcare systems because it aids to analyze and classify different types of diseases. It employs advanced computer vision and machine learning (ML) techniques [1] to accurately detect and diagnose different types of diseases [2,3]. These advanced techniques analyze complex visual data from imaging modalities like X-rays, MR images, and CT scans. This capability improves the survival rate of patient by advancing rapid and effective treatment planning as well as increasing diagnostic precision. Medical professionals may focus their attention to patient care and less time to labor-intensive image interpretation by automating the classification

process [4]. Medical image classification improves the diagnostic accuracy and healthcare outcomes by automating image analysis. It will reduce the cognitive load on medical professionals and allowing for more timely decision-making [5]. Furthermore, DL techniques may excel the workflows, reducing the interpretation time and also improving the quality of treatment that is provided to patients [6]. Despite its advancements, this method continues to face limitations that may hinder the accuracy and efficiency of the classification process. Factors like, Lacking of training data, variability in images quality caused by the environment, and interpretability of models can impede the optimal performance in medical imaging [7,8]. Biases in dataset composition can cause inaccurate diagnoses, when models come across

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unseen datasets [9]. The usage of automated systems typically limits the interpretability of results, making validation difficult for physicians. Moreover, the adoption of ML models into clinical practice is often hindered by clinician approval [10]. Medical image classification uses a variety of specific imaging modalities to improve the diagnostic accuracy across a broad spectrum of medical conditions. For respiratory issues, chest x-rays and CT scans are critical for detecting lung opacity [11] and complications from viral infections such as COVID and viral pneumonia [12–14]. In case of neurological examinations, brain MRIs [15] are useful for diagnosing conditions such as glioma, pituitary and meningioma [16]. Similarly, endoscopic images are essential for gastrointestinal assessments, aiding in identifying diseases such as esophagitis and pyloric disorders [17]. These modalities are strategically chosen to offer detailed and useful visual information, crucial for early detection and efficient treatment planning.

Traditional medical image classification methods tend to stand from limitations such as failure to effectively capture more intricate patterns and dependence on manual feature extraction, which might yield discrepancies and reduced diagnostic accuracy [18]. AI addresses such deficiencies by automating feature extraction and employing diverse data representations. DL techniques [19,20], ensemble learning [21,22], and feature fusion strategies [23,24], including multi-scale feature fusion [23], are utilized to enhance the robustness and accuracy of medical image classification. The advantages of using DL in this context include the improved diagnostic precision, efficiency, and capability to effectively handle massive datasets, ultimately contributing to better patient outcomes and streamlined clinical workflows.

1.1. Contributions and novelty

Diverse medical diseases increasingly impact human health, demanding accurate early diagnosis to prevent extensive damage to the human immune system. DL-based models, particularly pre-trained CNN [22,25–28], play an important role in timely disease prediction. However, current solutions face challenges. Training models on single diseases can result in biased predictions when applied to other conditions, reducing their generalizability and increasing the risk of misdiagnoses. Additionally, models trained only single diseases may not effectively capture the complexity of co-occurring conditions, which compromises their utility in real-world clinical settings where multiple diseases frequently exist. In this paper, we introduce a novel model, the Multi-Disease Learning Method, which is capable of learning diverse features across multiple diseases with fewer parameters while maintaining higher accuracy compared to single-disease models. This proposed model not only detects a variety of diseases but also incorporates an optimally truncated structure called the MobileLightNet architecture. The original MobileNet, though efficient due to its lightweight structure, often compromises on accuracy for multi-disease diagnosis tasks due to its design primarily for single disease learning. We have truncated the original MobileNet architecture to explore the optimal compression point. Furthermore, to address the challenge of recognizing diverse disease features at various scales which can lead to less performance in detecting finer details or broader contextual features. We have incorporated a multi-scale feature learning block into the truncated MobileNet model. At the end of our study, we employed nature-inspired metaheuristic algorithms [29], specifically Chain Foraging and Cyclone Aging approaches, to optimize the truncated model structure within a multi-scale feature learning block. Our experiments demonstrate that the proposed model surpasses previous studies and outperforms single-model learning strategies. By utilizing multi-level feature extraction within an optimized lightweight architecture, our results indicate that the model not only achieves high accuracy but also maintains low computational costs, facilitating timely disease diagnosis.

The main contribution of this work as follows:

- In this study, we introduce a novel and efficient approach, termed One Model for Multiple Diseases, which replaces single-disease models with an amalgamated framework capable of diagnosing multiple diseases simultaneously. This approach incorporates advanced techniques such as model truncation, a modified naïve inception block, multi-scale feature learning, and optimization via metaheuristic methods. By compressing the model, we achieve a lightweight, multi-scale inception architecture that maintains high accuracy with fewer parameters, enhancing its capability in multi-disease detection.
- The proposed model employs the MobileNet architecture for compression, demonstrating high performance in preliminary experiments compared to its counterparts. Designed for efficiency, MobileNet boasts a lightweight structure with fewer parameters, which facilitates easier model truncation and reduced computational requirements. Utilizing MobileNet as the base model ensures a balance between high performance and low resource usage. This approach enables the creation of an optimal, resource-efficient MultiDisease learning model suitable for diagnosing a variety of medical conditions where resource constraints are a significant factor.
- Our proposed solution addresses the limited generalization capabilities across various resolutions or conditions, which often stem from models learning only at the training scale. This limitation can degrade performance in multi-disease diagnosis, where the size and scale of features significantly vary. By optimally truncating the MobileNet architecture and incorporating a modified naïve inception block, we enable multi-scale feature learning within the model layers. This approach not only allows the model to integrate features at diverse scales but also prevents the dying ReLU issue by maintaining a small, non-zero gradient even when the unit is inactive. These enhancements collectively improve the model accuracy and robustness, making it more effective in handling complex diagnostic tasks.
- Optimizing the truncated MSFL model structure enhances robustness and scalability, enabling better generalization for multi-scale feature learning. We employed the Chain Foraging and Cyclone Aging techniques to optimize a truncated MobileNet architecture, alongside a modified naïve inception block, ensuring optimal structural and hyperparameter configurations. To further enhance feature extraction across spatial and temporal dimensions, we integrated a ConvLSTM unit before the Softmax layer. This adjustment has significantly improved performance in multi disease diagnosis with varying feature sizes and scales.

In the following section, we will provide a detailed literature review on the recent advancements in medical image classification. Subsequently, we will outline the methods and materials employed in our proposed methodology. This will be followed by a description of the experimental setup and the results obtained. We will then discuss the research limitations identified during our study. Finally, we will conclude with insights and future directions for the proposed methodology.

2. Literature review

ML has revolutionized healthcare systems by advancing medical imaging to improve the disease diagnosis and treatment planning. DL, a subfield of ML [25], has further enhanced these capabilities by improving the precision and reliability in biomedical research. DL models facilitate the analysis of intricate imaging data by offering higher levels of precision and effectiveness than conventional ML methods. These models play a key role in recent medical research and imaging. Different methodologies for analyzing the medical images have been studied, such as Ekong et al. [30] introduced a Bayesian Depthwise CNN method for classifying brain tumors from MRI and achieved an

validation accuracy of 94.32 %. Despite its effectiveness, their approach is impeded to lower validation accuracy and limited generalization to diverse datasets. Apostolopoulos et al. [31] employed deep convolution neural network (DCNN) to diagnose tumors from brain MR images and reaches to an accuracy of 95.65 %. The authors achieved remarkable performance but their method is hindered by the limited diversity of dataset and suboptimal parameter tuning. Mohanty et al. [32] proposed a Soft Attention CNN model for the diagnoses of brain tumors using brain tumor images and accomplish to 95.10 % accuracy. Their model faced suboptimal CNN performance and limited generalizability across diverse datasets. Study [33] implemented a GAN-CNN approach for brain tumor categorization and attained 95.60 % accuracy on brain MRI dataset. The method is limited by a 64×64 input size due to GAN constraints, reducing the compatibility with advanced architectures and impeding the ability of model to capture finer image details. Another study [34] employed a CNN-based method for brain tumor classification using MR images, reaching an accuracy of 94.20 %. The study faced limitations like, limited dataset diversity and testing on a single modality image. These constraints could impact the model's ability to generalize on larger datasets. Nishio et al. [35] introduced a DCNN based method for lung cancer classification and achieved the validation accuracy of 68.00 %. The study encountered challenges such as lower accuracy and demand higher computational resources. It was also limited to using a single modality image, which could reduce its ability to generalize well on other datasets. Togacar et al. [36] introduced a BrainMRNet model for detecting brain tumors. Their model integrated attention modules and utilized the hypercolumn technique to improve its performance. It achieved a high accuracy of 96.05 %, surpassing the results of commonly used pre-trained models in this domain. DL models have demonstrated effectiveness in COVID-19 detection by offering a variety of novel approaches. Researchers employed a ResNet50 model for COVID-19 detection [37]. They improved the model's performance by adding wavelet coefficients from the whole image to ResNet's fundamental framework and achieved a notable accuracy of 92.2 % for accurate diagnosis. Loey et al. [38] evaluated five deep CNNs for detecting COVID-19 samples. They improved the detection performance of these models by applying data augmentation using conditional generative adversarial network (CGAN). The ResNet50 model achieved diagnostic accuracy rate of 82.9 % among all five models. The study hindered lower accuracy along with higher computational demand due to deep architecture. Elmagzoub et al. [39] developed a framework for classifying gastrointestinal (GI) disorders based on the Kvasir dataset. They implemented seven transfer learning models to improve the detection of key features in endoscopic images. Following grid search optimization, the ResNet101 model achieved a 93.50 % accuracy. However, relying on a single model may limit generalizability and accuracy across diverse populations. Goyal et al. [40] developed the F-RNN-LSTM model for detecting lung diseases using chest X-ray images, specifically targeting COVID-19 and pneumonia. The model achieved 94.31 % accuracy on the CXIP dataset and 95.04 % on C19RD dataset. It performed well with low computational effort. However, it needs improvement to handle larger datasets with more classes and deeper models. Karthikeyan et al. [41] developed a CNN and Ranking algorithm to detect colorectal cancer, aiming of automating diagnostics and to alleviate pathologist workload. The model was evaluated on digital pathology datasets and achieved an accuracy rate of 91 %. However, its reliance on manual examination by pathologists may result in subjectivity and variability in diagnoses.

3. Methods and materials

This section presents a comprehensive description of the proposed (One Model for Multi Disease) model, designed for the detection of multiple diseases. Initially, we introduce various techniques, including feature extraction using CNN models, model truncation and compression, and multi-scale feature learning (MSFL). To enhance the MSFL

features, we incorporate ConvLSTM for sequential diverse feature learning. Furthermore, we optimize the proposed model using nature inspired techniques [29] such as Chain Foraging and Cyclone Aging, which contribute to the model robustness and efficiency in handling diverse disease indicators.

3.1. Proposed methodology

Previous studies [42,43] have shown that DL-based CNN models are of ongoing interest in the research community, particularly in the medical field. DCNN architectures can efficiently learn the intricate and complex hierarchical features of various diseases due to their deep architecture. In this work, we utilized the pretrained MobileNet model as our base model. Initial experimental setups demonstrate that MobileNet outperforms other models, achieving the highest results. MobileNet models achieve competitive accuracy with streamlined architectures, making them suitable for truncation where model complexity and processing power are limited. Moreover, these models are designed to maintain high accuracy even when scaled down, ensuring effectiveness across a wide range of medical diseases. After selecting the base model, we have divided the MobileNet architecture into three distinct segments. Each segment has been tested individually with optimal hyperparameter settings. This modular approach enhances debugging and validation by isolating components, simplifying error identification and resolution. It also allows for more effective resource allocation by identifying the most resource-efficient points for truncation, balancing computational costs with performance. Additionally, this method supports targeted fine-tuning of each segment, potentially improving overall model performance by focusing optimization efforts more strategically. In the subsequent phase, following the optimal truncation layers, we employ Multi-Scale Feature Learning. This approach significantly boosts model performance by leveraging diverse data representations, which enhances accuracy and robustness in predictions. Moreover, it augments the model generalizability across various disease features and conditions by capturing comprehensive interactions among features. Finally, we utilize nature-inspired approaches, such as Chain Foraging and Cyclone Aging, to determine the optimal structure for the multi-scale truncated model. Detailed descriptions of each step are provided below.

3.1.1. Feature extraction with pretrained CNN model

TL is a widely used ML technique that enhances the performance of DL models in medical imaging. This method involves transferring knowledge from a model pretrained on a large, often non-medical dataset to a related task, such as medical image analysis. By utilizing pretrained models, TL enables more efficient and effective learning in various domains where data may be infrequent or highly specific. Models can benefit from knowledge derived from large, diverse datasets in different domains, such as common images, by using TL. This approach offers the advantage of expedited training processes. Pre-trained models come equipped with a proficiency in generic features such as the detection of edges, shapes, and textures that can be applied across a range of tasks. Using pre-trained weights allows a model to start from an improved initialization point during training, which can accelerate convergence and reduce the training duration.

Previous research [44] has developed numerous advanced CNN models that outperform in various medical image classification tasks. However, challenges such as limited classification learning, specific disease dataset and constraints on computational resources often issue in progress. This research proposes a strategic approach to mitigate these issues by truncation, MSFL, and optimized truncated model that offer both high performance and cost-effectiveness.

3.1.1.1. MobileNet. The lightweight MobileNet architecture [45] is specifically designed for edge devices with limited computational

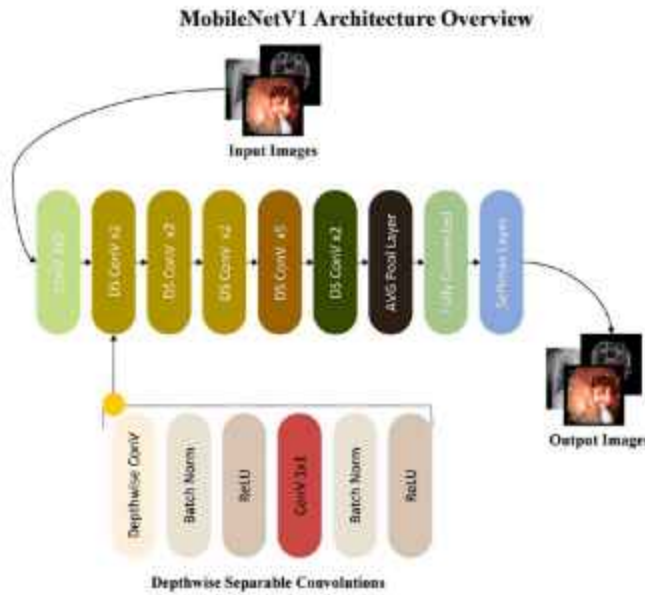


Fig. 1. Overview of MobileNet Architecture Diagram.

power, addressing the challenge of deploying DNN on hardware with restricted processing and memory capabilities. This innovative architecture is particularly beneficial for mobile and embedded systems, where resource constraints often hinder the performance of more complex models. MobileNet achieves its efficiency through the use of depthwise separable convolutions, a technique that decomposes the standard convolution operation into two simpler steps: depthwise convolution and pointwise convolution. This method significantly reduces the required computational resources and the number of parameters while maintaining robust image classification accuracy, allowing MobileNet to deliver high performance without losing speed. MobileNet design also incorporates several optimizations that enhance its applicability in various medical real-world scenarios. For instance, it supports adaptable width and resolution multipliers, enabling users to scale the model according to their specific requirements for accuracy and efficiency. This flexibility makes MobileNet particularly optimal for a wide range of applications, including real-time object detection, and image classification, especially on resource-constrained smartphones and IoT devices. Moreover, the architecture has undergone various iterations, such as MobileNetV2 and MobileNetV3, which introduce improvements in speed, accuracy, and feature representation. These advancements further prove MobileNet model as an efficient model choice in the field of computer vision, effectively bridging the gap between performance and resource efficiency. Fig. 1 presents the overall architecture of the MobileNet model.

3.2. Compression and truncation

This section details the model truncation and compression procedures. The MobileNet model has chosen for its balance between complexity and performance. In the truncation process, we removed the DS layers from bottom to top in order to reduce the number of model parameters while maintaining high accuracy across diverse disease classifications. Throughout this process, the original model design remained intact. Previous studies [46] have shown that the truncation process does not reduce performance or introduce overfitting, even with small datasets. To enhance performance with reduced computational costs, this study initially segmented the base model into three truncated sections: Start, Mid, and End. This approach involved removing the MobileNet classifier and several blocks, significantly reducing the number of trainable parameters.

- Start Truncation:** In the truncation process, DS blocks were sequentially removed from the MobileNet architecture to assess their impact on model performance. Originally, MobileNet features six DS blocks in the Start segment (Fig. 2(a) Original Architecture): DS₃₂ (-6), DS₆₄ (-5), DS₁₂₈ (-4), DS₁₂₈ (-3), DS₂₅₆ (-2), and DS₂₅₆ (-1). The truncation followed a bottom-up approach, where each DS block has removed one by one. Initially, the DS₂₅₆ (-1) block has removed from the bottom of the model, and DS₂₅₆ (-2) connected directly to the Mid truncation segment as shows in Fig. 2(b). Subsequently, the truncated model has constructed, trained, and evaluated for performance. If the truncated model demonstrated significantly improved accuracy compared to the original model, this configuration has selected as the optimal compressed architecture. if the performance gain has not sufficient, further truncation has performed by removing two sequential DS blocks, such as DS₂₅₆ (-2) and DS₂₅₆ (-1), and connecting DS₁₂₈ (-3) with the Mid truncation segment. This iterative process continued until no further performance improvement has observed for the Start truncation.
- Mid Truncation:** The same truncation approach has applied to the subsequent phases, specifically in the Mid truncation segment as illustrated in Fig. 2(c). This phase featured five identical DS blocks: DS₅₁₂ (-5), DS₅₁₂ (-4), DS₅₁₂ (-3), DS₅₁₂ (-2), and DS₅₁₂ (-1). The procedure involved sequentially removing DS blocks from the bottom of the Mid segment to evaluate the effect on model performance. Initially, one DS₅₁₂ block at a time has removed, and the modified model has tested to determine if performance improved or not. This systematic method ensured that the Mid truncation phase has optimized for both efficiency and effectiveness.
- End Truncation:** The End truncation (Fig. 2(d)) involved two DS blocks: DS₅₁₂ (-2), and DS₁₀₂₄ (-5).

Each truncation has performed independently to freeze the remaining sections of the model, thereby separating the effects of specific truncations. This method demonstrated a substantial reduction in

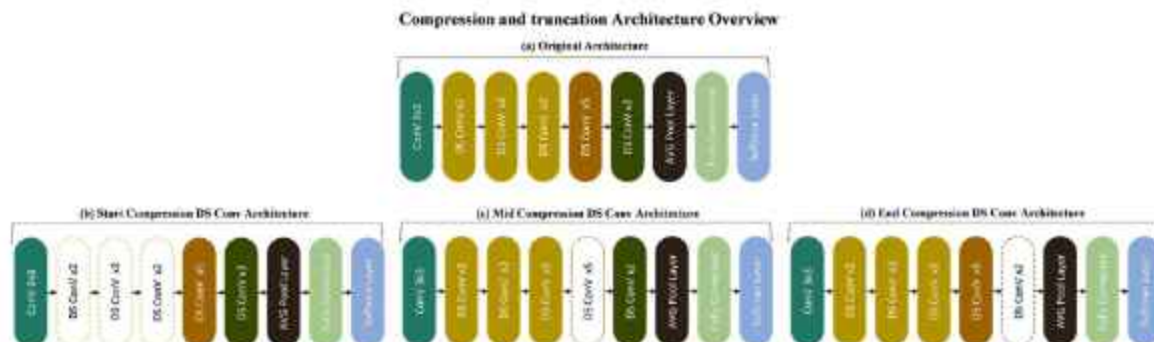


Fig. 2. Overview of Model Compression and truncation Diagram: (a) Original Architecture, (b) Start Compression, (c) Mid Compression, and (d) End Compression.

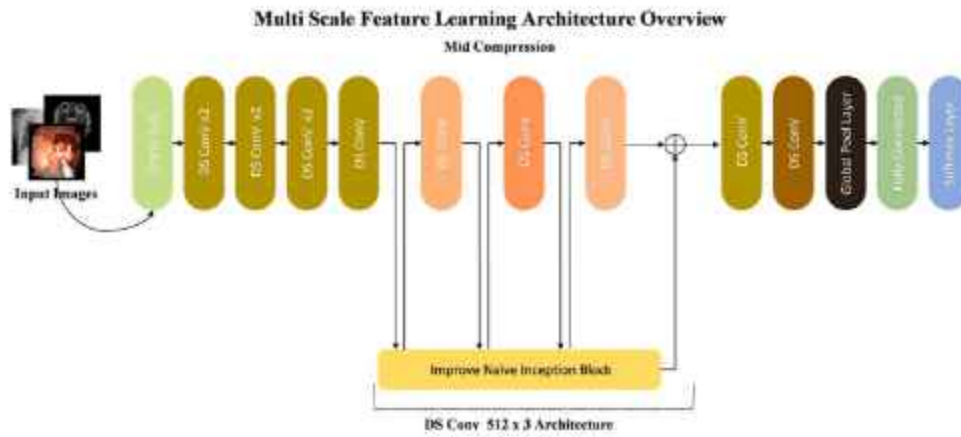


Fig. 3. Multi-Scale Feature Learning Architecture Overview.

parameters compared to the original model. Fig. 2 illustrates the truncation approaches between the original MobileNet and its truncated version. Before truncation, the original MobileNet model comprises 3.68 million parameters, which is reduced to 2.16 million parameters after truncation. The MobileNetV1 architecture initially features a sequence of 28 layers, including convolutional layers and depthwise separable convolutions, organized into 13 blocks. Each block typically contains DS layers, consisting of a depthwise convolution followed by a pointwise convolution (1×1 convolution). Following truncation, the model retains only 11 blocks, leading to a notable decrease in the total number of parameters and a simplified structure.

3.3. Multi-scale feature module

This section presents a detailed description of the utilization of a modified naive inception block in a truncated MobileNet model for multi-scale feature learning. This allows models [47] to capture and integrate details from various scales and resolutions, enhancing their ability to recognize patterns both in large, global structures and fine, local details. This approach improves the model's generalization capabilities across different contexts of medical image and datasets, leading to more robust performance. In this study, we have employed modified naive Inception module, within truncated model as a multi-branch structure, allows for the parallel processing of input data through different types of filters. This enables the model to capture a variety of features from the input data at multiple scales, enhancing the model ability to generalize across different disease patterns and scales. In contrary, failing to capture multi-scale features during model training can compromise the detection of finer details in smaller objects, reducing overall accuracy. Moreover, this limitation can hinder the model's robustness to scale variations, affecting its performance in diverse application environments. Fig. 3 presents the multi-Scale learning architecture overview.

3.3.1. Enhancing the basic inception block: an improved strategy

The MSFL architecture uses multiple scales, which allows for parallel processing and extraction of features at various levels of feature scales. Inception blocks [48], known for their ability to handle data at multiple scales simultaneously, are integral to this architecture. They use various sizes of filters within the same layer to capture different aspects of the image data. By incorporating Inception blocks, the model can effectively capture features at multiple scales, enhancing its ability to recognize patterns both in small and large spatial extents within the image. This architectural design aids in recognizing finer details while retaining contextual information from larger areas, improving the overall performance of the model on diverse medical disease diagnosis classification. In this study, we have improved the original inception block

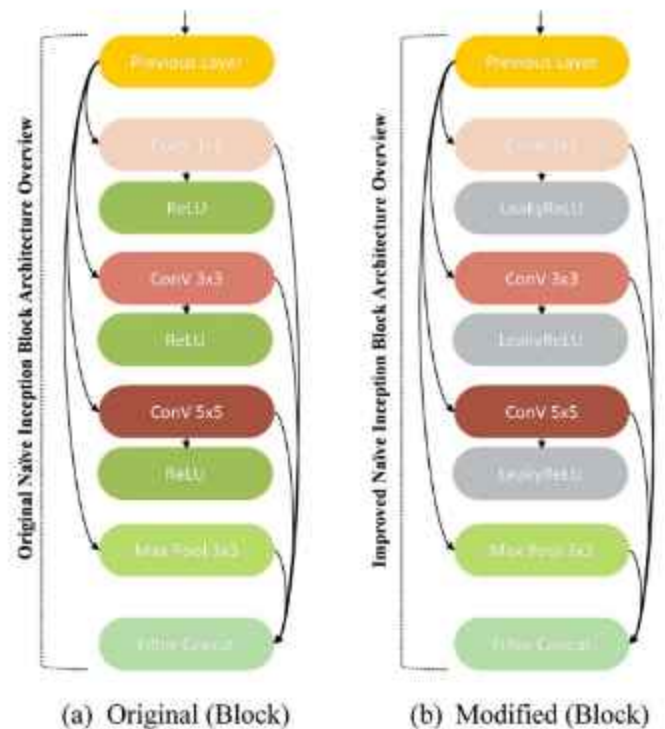


Fig. 4. Overview of Inception Block: (a) Original (Block), and (b) Modified (Block).

structure because block can lead to the dying ReLU issue, where neurons permanently deactivate and stop contributing to the learning process if they consistently receive negative inputs. This problem can cause issue in complex architectures like the Inception block, as it may result in significant feature loss within the network. To handle this, we have utilized the LeakyReLU layers within original inception block which allows a small and positive gradient when the neurons is inactive, promoting a sufficient gradient flow throughout the network. Fig. 4 presents the Overview of Inception Block: (a) Original (Block), and (b) Modified (Improved Strategy Block).

3.4. Boosting multi-scale feature with ConvLSTM

In this section, we have presented the details discussion about improving MSFL generated feature by using Long Short-Term Memory (ConvLSTM). This is a type of RNN optimized to recognize and remember feature patterns in multi-images over long periods. This

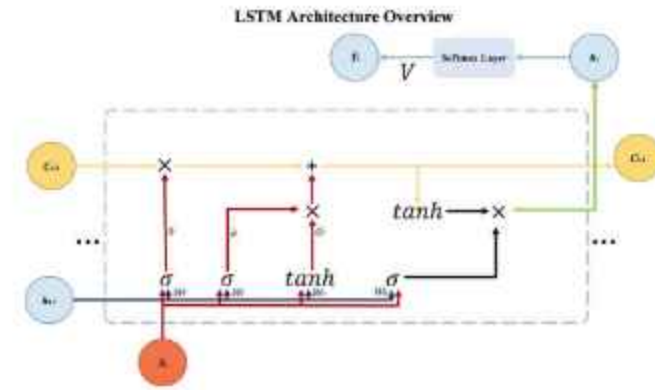


Fig. 5. Overview of LSTM UNIT architecture diagram.

capability makes ConvLSTM ideal for monitoring and modeling multiple disease growth, allowing for a deep understanding of cell evolution and response to treatments [49]. They use a gating mechanism to selectively retain and discard feature, which is crucial for modeling the complex patterns of disease progression. ConvLSTM effectively handle the variability in disease growth, influenced by various factors such as treatments and genetic changes [50]. They analyze historical feature to predict future growth trajectories and accommodate multiple data types, including gene expression and imaging data. ConvLSTM provide a comprehensive tool for advancing cancer research and developing personalized therapies, by capturing and analyzing the temporal dynamics of disease feature growth [51–53]. Fig. 5 presents the basic architecture overview of ConvLSTM unit.

This study has utilized the ConvLSTM after the MSFL module to further refine the representation of each disease features before passing them to the softmax layer. By integrating ConvLSTM, the model captures temporal dependencies within the image feature, enhancing the robustness and accuracy of the predictions. This sequential processing ensures that the softmax layer receives well-structured inputs, optimizing the classification performance.

3.5. Optimizing truncated MobileNet architecture

Optimizing [54] the entire model structure can lead to more fundamental improvements in performance and efficiency, enabling the model to better handle complex patterns and scale effectively with larger datasets. such as integrating, new layers, adjusting the connectivity or compressing model structure, can address limitations in model capacity and generalize better across diverse disease. In contrary to it, only hyperparameter tuning might not achieve it. A comprehensive optimization of the model architecture allows for a more efficient approach to enhancing model robustness and interpretability. Previous studies show that optimizing model hyperparameters using nature-inspired algorithms for medical disease diagnosis can enhance model performance by effectively exploring and exploiting the search space to find optimal configurations model structure. These algorithms can adaptively adjust parameters, leading to improved accuracy and robustness in diagnosing diseases. Additionally, these algorithms often require fewer computational resources compared to traditional grid search methods, making them cost-effective for large-scale medical applications. In this work, we have used two approaches namely the Chain Foraging and Cyclone Aging, to optimized the truncated MSFL architecture for multiple disease diagnosis. In this optimization process, our research follows a systematic approach. Initially, we generate a population of three potential solutions, each representing a unique set of kernel sizes. We then enhance these solutions through four iterative rounds, with each round comprising 30 epochs. This structured process allows for the gradual optimization of our truncated MSFL model performance.

3.5.1. Objective function

To enhance the accuracy of diagnosing multiple diseases, we refined our model structure using Chain Foraging and Cyclone Aging techniques. These methods iteratively optimized the model structure by evaluating each population against standard assessment criteria over a set number of iterations. Specifically, the optimizer adjusted the kernel values of MDL_{nk} during this process. This approach efficiently reduced the search space, often achieving an optimal model structure within a maximum of four iterations. If no further improvements were observed, we concluded that the model configurations at that stage were optimal for addressing the search problem. The fitness functions for the Chain Foraging and Cyclone Aging methods are detailed in Eq. (1) below.

Imagine a scenario where a solution is defined by a collection of distinct kernel values, denoted as MDL_{nk} . In each iteration x of the population, these kernel values are dynamically adjusted to optimize the search for the highest accuracy in the model. The model undergoes training using validation data, and its performance is evaluated using metrics such as accuracy, precision, recall, and f1-Score. The best configuration value of MDL_{nk} and the structure of the model are recorded based on the top scores within each population iteration x . A cumulated score is computed using these metrics. The objective function aims to find the minimum error rate value with maximum validation accuracy.

$$\text{Objective} = \text{Error}_{\text{Rate}} = 1 - \text{Valid}_{\text{Acc}}$$

$$MDL_{nk} = \text{kernel size}(3, 5, 7, 11) \quad (1)$$

3.6. Proposed architecture overview

This section provides a comprehensive description of the proposed framework designed to enhance medical image diagnosis through a TL based approach. The framework implements a multi-disease diagnostic strategy that starts with image preprocessing and augmentation steps. These preliminary processes aim to improve image quality and variability, ensuring that the subsequent diagnostic model can accurately identify and classify multiple diseases from medical images. Image processing techniques like augmentation enhance model robustness by generating diverse training examples from existing data, improving generalization. They also help reduce overfitting by introducing variability, making the model more adaptable to different scenarios. In the subsequent phase, we have employed three truncation, and model compression approach namely model truncate at the beginning, at the Mid, and end model truncation. Reducing model parameters is essential to enhance computational efficiency and reduce overfitting while maintaining high performance levels. Multi-scale feature learning is crucial in truncated models as it enhances the ability to capture and integrate various medical disease features from different levels of granularity, improving overall model performance and robustness. To enhance the model feature learning capabilities, we have integrated the MSFL module into the truncated model. The incorporation of MSFL thus facilitates a more complex and comprehensive feature representation. In the proposed methodology, the next step involves optimizing the truncated MobileNet model structure through the two nature-inspired approaches. The Chain Foraging approach adjust model structures (Kernel Size) by sequentially identifying and integrating optimal features through iterative refinement. Cyclone Aging optimizes model structures by progressively updating and refining parameters, leveraging temporal information to improve accuracy and robustness. Together, these methods fine-tune models for better performance and adaptability. Different diseases exhibit a wide range of features that can vary significantly in terms of appearance, and distribution. This variability can make it challenging for model to learn and generalize these features effectively. To address this, we incorporated an ConvLSTM layer prior to the softmax layer in our model framework. ConvLSTM layer excel in capturing long-term dependencies in sequential image features,

enhancing performance in tasks requiring memory of past feature. By taking use of its special capacity to handle sequential image features, the ConvLSTM unit integration prior to the Softmax layer significantly improves feature extraction across spatial and temporal dimensions, especially for changing feature sizes and scales. Since ConvLSTM units are made to hold onto information over time, the model may identify

different sequences by integrating the ConvLSTM unit. Fig. 6 presents the proposed methodology framework diagram.

In this section, we have described (Algorithm 1) the overall steps and procedure of One model for multiple disease detection framework.

Algorithm 1. Multiple Medical Disease Detection Model

Input: Multiple Medical Disease Image

1. Create variable W, H, C to defined the image dimension
2. Standardized the Input image sizes for MobileNet model such as (224 x 224 x 3)
3. Apply Image processing, and Augmentation steps e.g., Rescale, Flips, Rotation
4. Define a sequential model structure layer (MobileNet)
5. Begin Model Truncation and Compression Button to Top approach (original model structure remain same)

Start Truncation- DS_S $\because$ Depthwise Separable Block (6) Remove One by One

Mid Truncation- DS_M $\because$ Depthwise Separable Block (5) Remove One by One

End Truncation- DS_E $\because$ Depthwise Separable Block (2) Remove One by One

6. Multi-Scale Feature Module fuse within Truncated Model
Integrate Modified Naïve Inception block
7. Employ LSTM unit Before Softmax layer in Truncated MSFL Model
8. Optimized Model Structure

- Chain Foraging approach

$$M_k^r(x+1) = \begin{cases} M_k^r(x) + r \times (M_{best}^r(x) - M_k^r(x) + v \times (M_{best}^r(x) - M_k^r(x))) & k = 1 \\ M_k^r(x) + r \times (M_{k-1}^r(x) - M_k^r(x) + v \times (M_{best}^r(x) - M_k^r(x))) & k = 2, \dots, N \end{cases} \quad (2)$$

$$v = 2 \times r \times \sqrt{|\log(r)|} \quad (3)$$

- Cyclone Aging approach

$$M_k(x+1) = M_{best} + r \times (M_{k-1}(x) - M_k(x)) + e^{mr} \times \sin(2\pi r) \times (M_{best} - M_k(x)) \quad (4)$$

$$D_k^r(x+1) = \begin{cases} D_{best}^r(x) + r \times (D_{best}^r(x) - D_k^r(x) + \gamma \times (D_{best}^r(x) - D_k^r(x))) & k = 1 \\ D_{best}^r(x) + r \times (D_{k-1}^r(x) - D_k^r(x) + \gamma \times (D_{best}^r(x) - D_k^r(x))) & k = 2, \dots, N \end{cases} \quad (5)$$

$$\gamma = 2 e^{r_1 \frac{T-d+1}{T}} \times \sin(2\pi r_1) \quad (6)$$

$$D_{random}^{dr} = LL^{dr} + r \times (UL^{dr} - LL^{dr}) \quad (7)$$

$$D_k^{dr}(x+1) = \begin{cases} D_{random}^r(x) + r \times (D_{random}^r(x) - D_k^r(x) + \gamma \times (D_{random}^r(x) - D_k^r(x))) & x = 1 \\ D_{random}^r(x) + r \times (D_{k-1}^r(x) - D_k^r(x) + \gamma \times (D_{random}^r(x) - m_k^r(x))) & x = 2, \dots, N \end{cases} \quad (8)$$

Objective Function

$$\text{Objective} = \text{Error}_{\text{Rate}} = 1 - \text{Valid}_{\text{Acc}}$$

$$\because \text{MDL}_{nk} = \text{Kernal size } (3, 5, 7, 11)$$

9. Training Model with preprocessing, and augmented Multiple Medical disease images
 10. Evaluate the Proposed Model using test dataset
 11. Finalized results, and model weights sane for future perspective
-

temporal dependencies in the features of the image. Because of variations in scale, orientation, or context, the same features may appear differently in different slices of medical imaging. Our suggested framework can efficiently represent the correlations between temporal progression of spatial data acquired from convolutional layers across

4. Experimental setup and results

This section provides a comprehensive overview of the experimental setup, including assessment metrics, dataset details, preprocessing, augmentation methods, model training hyperparameters, and the results

Overview of Proposed Methodology Framework

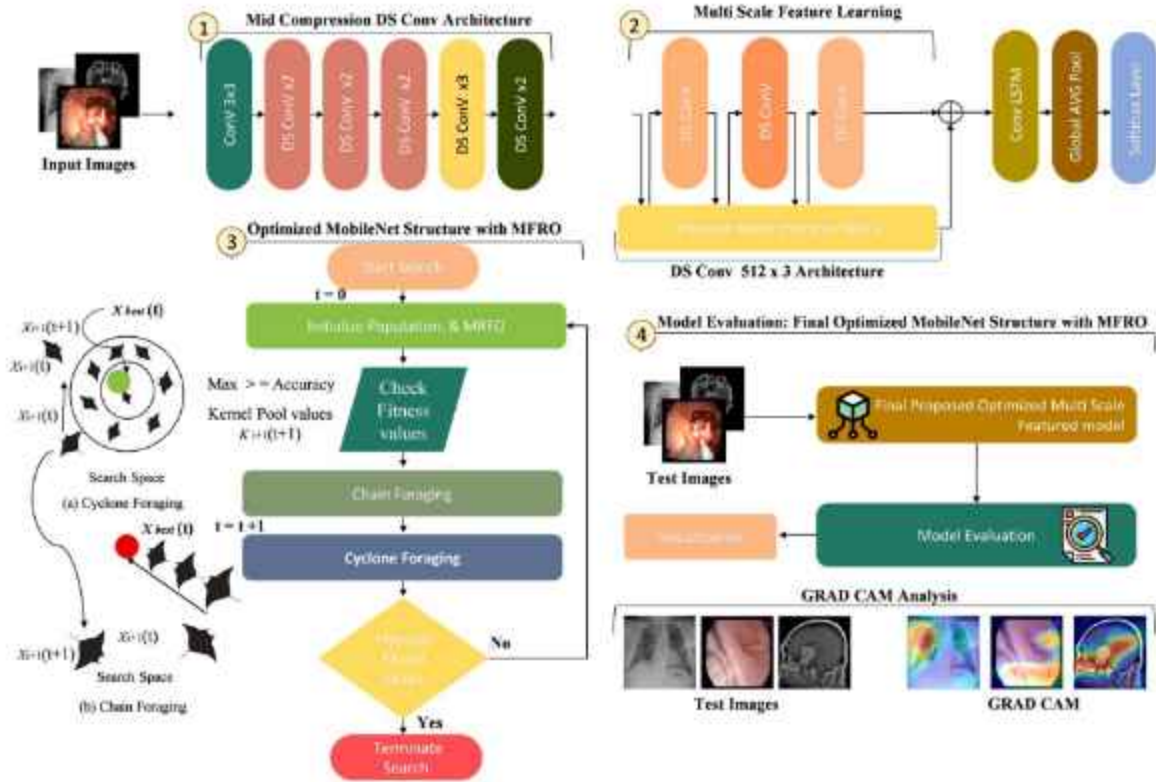


Fig. 6. Overview of Proposed (Our) methodology: One model for multiple disease.

Table 1
Overview of Dataset Distribution.

Class	Train	Test	Validation	Total
Colorectal cancer	100	28	11	139
COVID	1440	400	160	2000
Esophagitis	77	21	9	107
glioma	1189	300	132	1620
Lung Opacity	1440	400	160	2000
meningioma	1206	306	133	1645
pituitary	1311	300	146	1757
Pylorus	108	30	12	150
Viral Pneumonia	968	269	108	1345

achieved by our proposed model on various medical image datasets. To evaluate the efficacy of our model, we compare its performance with results from different pre-trained CNN architectures and current techniques, employing various blocks and methodologies. Additionally, we present an in-depth ablation study to validate the contribution of each technique within the proposed model, demonstrating the effectiveness of our approach.

4.1. Metrics to evaluate model performance

In this study, we utilized various statistical evaluation metrics accuracy (Ac), precision (Pr), recall (Re), and F1-score (F1) to assess the robustness and efficacy of the proposed model. These metrics are derived from the confusion matrix, which includes True Positives (P_{true}), True Negatives (N_{true}), False Positives (P_{false}), and False Negatives (N_{false}).

$$Accuracy(Ac) = \frac{P_{true} + N_{true}}{P_{true} + P_{false} + N_{true} + N_{false}} \quad (9)$$

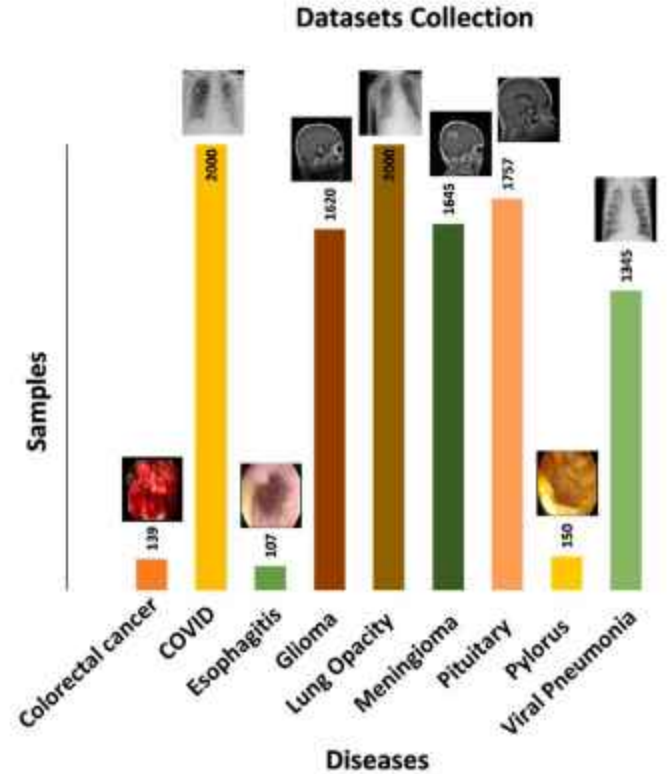


Fig. 7. Sample images for each disease in the dataset.

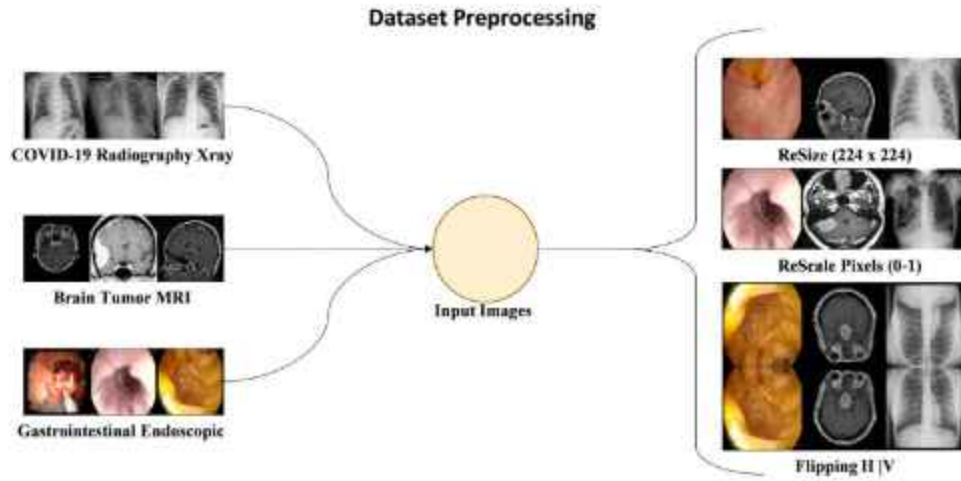


Fig. 8. Overview of Employment Augmentation Techniques.

$$\text{Precision(Pr)} = \frac{P_{true}}{P_{true} + P_{false}} \quad (10)$$

$$\text{Recall(Re)} = \frac{P_{true}}{P_{true} + N_{false}} \quad (11)$$

$$\text{F1Score(F1)} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (12)$$

4.2. Datasets collection

In this section, we provide detailed information about the datasets used to assess the reliability and efficacy of our proposed model in differentiating various medical conditions across diverse modalities. The datasets utilized in this study include MRI [42], Chest X-ray [55], and endoscopic [56] images. Each modality presents unique characteristics, such as infection regions and tumor sizes, which are crucial for evaluating the model performance across different medical scenarios. Table 1 provides an overview of dataset distribution across different classes for training, testing, validation, and total dataset size. For each class, the number of samples is divided into training, testing, and validation subsets. For example, the COVID class has 1440 samples for training, 400 for testing, and 160 for validation, totaling 2000 samples. Other classes, such as Colorectal cancer and Pylorus, have fewer samples, with totals of 139 and 150, respectively. The distribution varies significantly among classes, with some like Glioma and Meningioma having larger datasets, indicating a focus on balancing the dataset size across various medical conditions to ensure comprehensive model training and evaluation.

While the Fig. 7 presents the sample images for each dataset disease. It is stated in previous research [57,58] that MRI images play a crucial role in detecting brain tumors by providing detailed and high-resolution images of brain structures, allowing for accurate identification and localization of tumors. In this study, we have employed the Figshare benchmark dataset [42] to validate the robustness of proposed model. The dataset comprises MRI scans of three types of brain tumors: glioma, meningioma, and pituitary tumors. Table 1 provides the distribution of samples for each tumor type. Chest X-ray [55] are utilized to detect COVID, Lung Opacity, and Viral Pneumonia in patients by identifying characteristic lung abnormalities associated with the infections. The Chest X-ray dataset, which employ in this study contain total of 5345 images. Endoscopic images [56] are utilized to detect infections in patients with gastrointestinal conditions such as Esophagitis, Colorectal cancer, and Pylorus. In this work, we have assessed the classification performance of our proposed model using a publicly available set of GI endoscopic images. Table 1 presents the details of the sample

Table 2

Class wise Performance Analysis of MobileNet, Truncated Model, MSFL, and Proposed model.

Models	Class	Precision	Recall	F1-score	Accuracy
MobileNetV1-Base	Colorectal Cancer	1.00	1.00	1.00	95.32
	COVID	0.94	0.95	0.95	
	Esophagitis	0.95	0.86	0.90	
	Glioma	0.97	0.95	0.94	
	Lung Opacity	0.96	0.96	0.95	
	Meningioma	0.94	0.95	0.96	
	Pituitary	0.98	0.99	0.99	
	Pylorus	0.91	0.97	0.94	
	Viral	1.00	0.98	0.99	
	Pneumonia				
	Colorectal Cancer	1.00	0.96	0.98	95.95
	COVID	0.94	0.98	0.96	
Truncated Model (Mid)	Esophagitis	0.84	0.76	0.80	
	Glioma	0.96	0.92	0.94	
	Lung Opacity	0.98	0.95	0.97	
	Meningioma	0.93	0.94	0.93	
	Pituitary	0.98	1.00	0.99	
	Pylorus	0.82	0.90	0.86	
	Viral	1.00	0.99	0.99	
	Pneumonia				
	Colorectal Cancer	1.00	0.96	0.98	96.34
	COVID	0.96	0.97	0.97	
	Esophagitis	0.80	0.76	0.78	
MSFL Model	Glioma	0.97	0.95	0.96	
	Lung Opacity	0.97	0.96	0.97	
	Meningioma	0.95	0.86	0.90	
	Pituitary	0.97	0.99	0.98	
	Pylorus	0.81	0.87	0.84	
	Viral	1.00	0.99	0.99	
	Pneumonia				
	Colorectal Cancer	1.00	1.00	1.00	97.37
	COVID	1.00	0.98	0.98	
	Esophagitis	0.95	0.86	0.90	
	Glioma	0.98	0.96	0.98	
	Lung Opacity	0.97	0.98	0.98	
	Meningioma	0.94	0.95	0.94	
	Pituitary	0.97	0.99	0.98	
	Pylorus	0.82	0.90	0.86	
	Viral	1.00	0.98	0.99	
	Pneumonia				

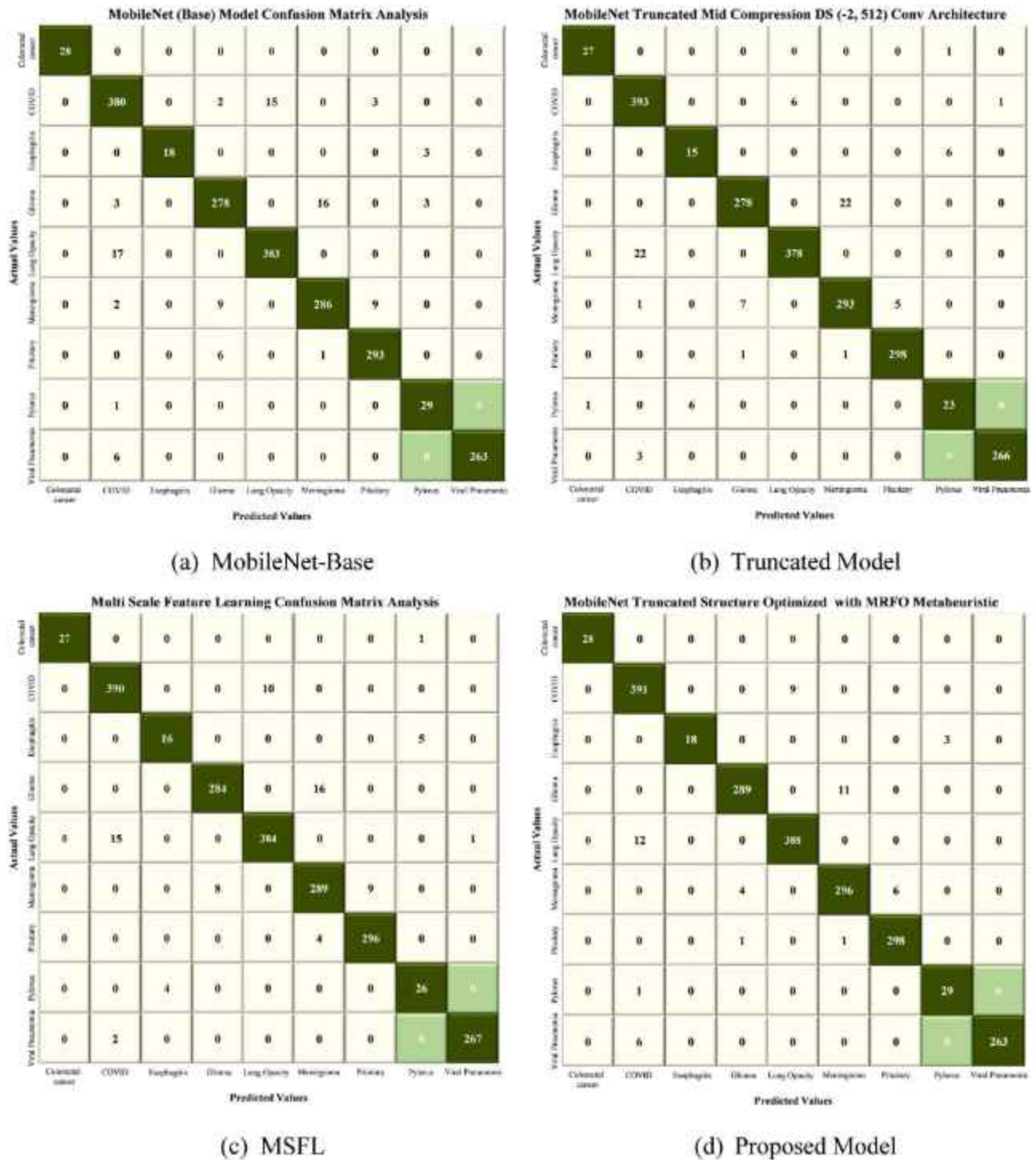


Fig. 9. Confusion matrix analysis: (a) MobileNet, Truncated Model (Mid), MSFL, and Proposed model.

distribution for training, validation, and testing across each class.

4.3. Preprocessing and augmentation details

In medical imaging [43], rescaling ensures that all images have consistent dimensions and pixel values, which improves model training stability and convergence. Augmentation techniques, such as rotations and flips, simulate variations in imaging conditions, helping to prevent overfitting and enabling the model to better generalize to diverse medical multiple disease scenarios. In this study, initially utilize medical

dataset have various in dimension size. Different class dimensions in a dataset can result in uneven learning and poor generalization, as model may become biased toward classes with larger dimensions. To address this issue, we first resize all the images to dimensions (224×224) suitable for the subsequent processing steps. Augmentation enhances model robustness by expanding the training data, leading to improved generalization and performance on unseen data. We have employed the three techniques such as rotation in augmentation enhances the robustness of proposed model by enabling it to generalize better across different orientations and angles. Secondly, we have utilized the Flipping which

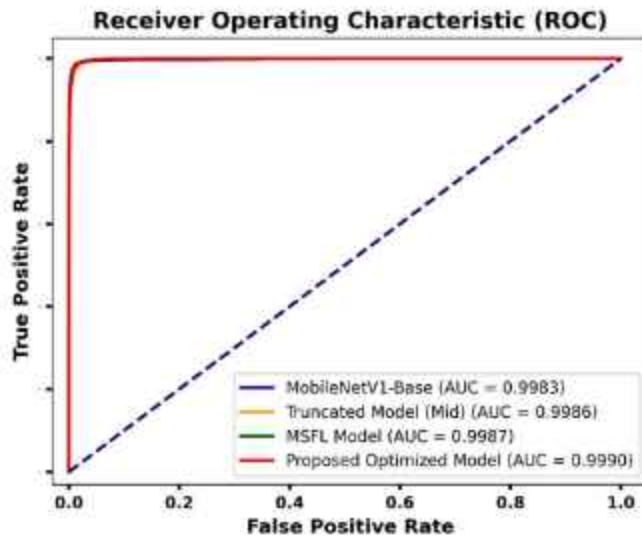


Fig. 10. Performance Analysis with ROC Curve.

increasing dataset diversity and reducing model overfitting. Fig. 8 presents the overview of multiple medical disease image augmented samples.

4.4. Classification performance

The proposed model shows high performance metrics across several classes. For instance, in the category of Colorectal Cancer, all models show better precision, recall, and F1-scores, but the proposed model leads with the highest overall accuracy at 97.37 %, compared to 95.32 % for MobileNetV1-Base and 95.95 % for the Truncated Model (Mid). Similarly, in categories like COVID, Glioma, and Lung Opacity, the proposed model either equals or outperforms the other models in precision and recall, leading to higher F1-scores. The proposed model maintains consistency in high scores across multiple categories, notably achieving consistent scores in the Viral Pneumonia detection as other models achieved. Table 2 provides a detailed class-wise performance analysis of MobileNetV1-Base, Truncated Model (Mid), MSFL Model, and the Proposed Model across various medical diseases. The evaluation metrics include Precision, Recall, F1-score, and overall Accuracy for each model, which are crucial for evaluating the effectiveness of the proposed model with its counterpart model in multiple disease image classification. The diseases analyzed include Colorectal Cancer, COVID, Esophagitis, Glioma, Lung Opacity, Meningioma, Pituitary, Pylorus, and Viral Pneumonia. The integration of the ConvLSTM unit before the Softmax layer significantly enhances feature extraction across spatial and temporal dimensions, particularly for varying feature sizes and scales, by leveraging its unique ability to process sequential image features. ConvLSTM units are designed to retain information over time, allowing the model to capture temporal dependencies in the image feature. This capability is particularly beneficial for medical imaging, where the same features may appear differently across various frames or slices due to changes in scale, orientation, or context. By incorporating the ConvLSTM unit, our proposed framework can effectively model the relationships between spatial features extracted from convolutional layers and their temporal progression across different sequences.

Fig. 9 presents the confusion matrix analysis between the proposed model and its counterparts, evaluated on a dataset of 2054 test samples. The MobileNetV1-based model, while showing impressive results due to its effective balance between computation and accuracy, misclassified 96 images. In contrast, the truncated model demonstrates superior performance relative to the base model, attributed to its optimal compression. Moreover, the incorporation of the MSFL with the truncated model marginally enhances performance compared to both the

Table 3

MobileNetV1-Base Model (Different Configuration) Start compression and truncation.

Models	Accuracy	Precision	Recall	F1-score
MobileNetV1DS (-1)	94.69	91.08	89.96	90.11
MobileNetV1 DS (-2)	94.88	92.35	91.79	91.96

Table 4

MobileNetV1-Base Model (Different Configuration) Mid compression and truncation.

Models	Accuracy	Precision	Recall	F1-score
MobileNetV1 DS (-1)	94.20	90.58	90.01	90.08
MobileNetV1 DS (-2)	95.95	93.81	93.35	93.52
MobileNetV1 DS (-3)	95.03	91.83	91.86	91.78
MobileNetV1 DS (-4)	94.83	92.25	90.44	90.78

base and truncated models alone. The proposed model, optimized through Chain Foraging and Cyclone Aging techniques within a truncated MobileNet architecture, outperforms all others by misclassifying only 54 images.

The ROC curve is vital for assessing model performance as it visualizes the trade-off between true positive and false positive rates, enabling a clear evaluation of classification effectiveness across different thresholds. Fig. 10 presents the ROC curve analysis of the proposed optimized model alongside other models. The proposed model achieved the highest AUC value of 99.90, outperforming the multi-scale learning (MSFL) model, which obtained an AUC of 99.87, and the optimal mid-truncated model, which achieved an AUC of 99.86. In comparison, the MobileNetV1-Base model attained an AUC value of 99.83, demonstrating lower performance relative to the proposed optimized model.

4.5. Compression and truncation results analysis

The truncated architecture plays a crucial role in balancing the trade-off between accuracy and computational cost in real-time medical clinical settings by reducing the model complexity without significantly compromising performance. By strategically removing less impactful layers, such as in the MobileNetV1 DS (-2) configuration, the model maintains high accuracy while decreasing the number of operations required. This reduction in depth allows the model to perform faster inferences, which is essential for real-time applications, where timely and accurate diagnoses are critical, particularly in resource-constrained environments like mobile devices or edge computing systems used in clinical settings. Table 3 presents the performance of the MobileNetV1-Base model when truncated at the start with different settings of a depthwise separable (DS) layer compression. The first compression, where from the last DS layer of the MobileNetV1 DS (-1) truncated, shows an accuracy of 94.69 %, precision of 91.08 %, recall of 89.96 %, and an F1-score of 90.11 %. The second configuration, where two last DS layers remove (MobileNetV1 DS (-2)), demonstrates slightly better overall performance with an accuracy of 94.88 %, precision of 92.35 %, recall of 91.79 %, and an F1-score of 91.96 %. This indicates that the DS (-2) setting yields improvements across all metrics compared to DS (-1). The increased precision and recall in the second configuration suggest that it is better at correctly identifying relevant disease and also reduces the number of false negatives, leading to a higher overall F1-score,

Table 5

MobileNetV1-Base Model (Different Configuration) End compression and truncation.

Models	Accuracy	Precision	Recall	F1-score
MobileNetV1 DS (-1)	95.76	93.19	92.88	92.99
MobileNetV1 DS (-2)	95.22	91.42	90.82	91.05

Table 6
Blocks wise testing of MobileNetV1 DS (-2) Architecture.

Models	Accuracy	Precision	Recall	F1-score
Residual Block	95.37	91.54	90.29	90.48
Channel Attention	96.05	93.31	92.96	93.08
CBAM	94.83	93.14	92.40	92.60
Dense Block	96.00	93.33	93.02	93.12
Inception Block	95.86	92.95	92.48	92.63
Naïve Inception Reduction Block	95.08	92.23	91.41	91.64
Squeeze and Excitation Block	94.88	92.91	92.40	92.57
Improved Inception Block	96.34	93.55	93.31	93.41

which balances precision and recall.

Table 4 presents performance metrics for different configurations of the MobileNetV1 model, specifically with mid-level compression and truncation, where depthwise separable denoted by DS followed by a configuration index. The configuration MobileNetV1 DS (-2) showcases the highest overall effectiveness with an accuracy of 95.95 %, precision of 93.81 %, recall of 93.35 %, and an F1-score of 93.52 %. In contrast, the configuration MobileNetV1 DS (-1) exhibits the lowest performance across all metrics, with an accuracy of 94.20 %, precision of 90.58 %, recall of 90.01 %, and an F1-score of 90.08 %. The configuration MobileNetV1 DS (-3) and MobileNetV1 DS (-4) fall in between, with DS (-3) slightly border out in terms of recall (91.86 % vs. 90.44 %) and DS (-4) having marginally better precision (92.25 % vs. 91.83 %) and F1-score (90.78 % vs. 91.78 %). Overall, each configuration shows high performance, but MobileNetV1 DS (-2) stands out as the most balanced and effective model compression and truncation.

Table 5 presents a comparison of the performance metrics for the MobileNetV1 Base model under different configurations, specifically with varying levels of end compression and truncation. The first configuration, MobileNetV1 DS (-1), shows higher overall performance across all metrics: an accuracy of 95.76 %, precision of 93.19 %, recall of 92.88 %, and an F1-score of 92.99 %. In contrast, the second configuration, MobileNetV1 DS (-2), exhibits slightly lower results with an accuracy of 95.22 %, precision of 91.42 %, recall of 90.82 %, and an F1-score of 91.05 %. The differences between these configurations highlight the impact of the degree of DS truncation, with the first configuration maintaining better overall performance, suggesting less information loss or more effective feature retention compared to the second.

The analysis of the MobileNetV1 model across different truncation and compression settings reveals that mid-level truncation generally results in better performance compared to both start and end truncation approaches. Specifically, the MobileNetV1 DS (-2) configuration consistently demonstrates the highest performance metrics across various tables, including the highest accuracy, precision, recall, and F1-score in both mid-level truncation scenarios and when compared to start and end truncation settings. For instance, Table 4 shows MobileNetV1 DS (-2) achieving an accuracy of 95.95 %, precision of 93.81 %, recall of 93.35 %, and an F1-score of 93.52 %, outperforming other configurations. In contrast, start truncation (MobileNetV1 DS (-1)) and end truncation (MobileNetV1 DS (-2)) exhibit lower overall performance metrics, with MobileNetV1 DS (-1) showing an accuracy of 94.69 % and MobileNetV1 DS (-2) with 95.22 %. These findings underscore that mid-level truncation, particularly MobileNetV1 DS (-2), provides a more balanced and effective model, preserving essential features better than start and end truncation methods and thus yielding improved performance across key evaluation metrics. The results have confirmed that the optimal truncation for the model is at the mid-level, and the last two depth-separation (DS) blocks at the bottom of the original mid-level MobileNet model are unnecessary. This has been validated by the findings.

Table 7
Performance Comparison of Proposed Model with pre-trained model Results.

Models	Accuracy	Precision	Recall	F1-score
MobileNetV1 (Base)	95.32	95.11	95.69	95.86
MobileNetV2 (Base)	95.13	93.01	92.63	92.68
MobileNetSV3 (Base)	83.15	80.08	81.74	81.12
MobileNetLV3 (Base)	85.00	82.16	81.45	81.77
DenseNet121 (Base)	92.44	92.62	93.75	92.40
DenseNet169 (Base)	91.47	90.67	91.39	90.63
DenseNet201 (Base)	90.83	89.25	89.64	89.18
InceptionV3 (Base)	87.72	88.44	89.29	88.02
ResNet50 (Base)	84.21	84.19	84.33	83.50
ResNet101 (Base)	84.41	81.35	82.41	81.21
VGG16 (Base)	87.72	88.44	89.29	88.02
VGG19 (Base)	87.99	87.36	88.09	86.76
Xception (Base)	85.79	84.88	83.98	81.96
Proposed (Our)	97.37	97.31	96.96	96.08

4.6. Performance comparison of proposed model with other blocks

In this section, we present performance validation of modified inception block through performance comparison with other blocks to evaluate the improvements in efficiency, accuracy, or feature extraction capabilities. It also highlights the advantages of the modifications, ensuring that the proposed changes are optimal for the multi disease prediction. Table 6 provides a comparative analysis of different blocks integration for multi-scale feature learning in the MobileNetV1 DS (-2) Mid truncated architecture based on various performance metrics. The improved Inception Block shows the highest overall performance, with an accuracy of 96.34 %, precision of 93.55 %, recall of 93.31 %, and F1-score of 93.41 %. Close counterpart includes the Dense Block and the Channel Attention Block, showing comparable accuracies around 96 % and F1-scores over 93 %. The Residual, Inception, and Squeeze and Excitation Blocks perform similarly, with accuracies ranging from 94.83 % to 95.37 % and F1-scores over 90 %. The Naïve Inception Reduction Block is the least effective among the evaluated blocks, with the lowest accuracy and F1-score of 95.08 % and 91.64 %, respectively. This indicates a varying degree of effectiveness in integrating complex features and channel dependencies across different block designs within the mid MobileNetV1 DS (-2) compression architecture enhance performance.

The results shows that Improved Inception Block emerges as the most effective, underscoring its efficient design in handling complex diverse features and channel dependencies. This block achieves the highest accuracy (96.34 %) and maintains impressive precision, recall, and F1-score metrics, suggesting a balanced enhancement across all evaluated performance dimensions.

4.7. Performance comparison of proposed model with pre-trained models

We conducted a thorough analysis by running existing pre-trained models under the same experimental settings, ensuring a fair comparison across different architectures. The results indicate that our proposed model not only matches but often surpasses the performance of these existing models in terms of accuracy, providing robust solutions for clinical applications. Evaluating the proposed model against pre-trained models is critical for demonstrating its efficiency by establishing a performance baseline using models that have been trained on large, diverse datasets. Pre-trained models offer well-optimized architectures, so outperforming them validates the proposed model ability to generalize effectively and leverage novel features. This comparison also clarifies the impact of the introduced modifications in architecture and optimization techniques, validating their effectiveness in improving performance. The performance comparison Table 7 below outlines the results of multiple pre-trained models with a proposed model, focusing on metrics such as accuracy, precision, recall, and F1-score. The proposed model outperforms all listed pre-trained models, achieving the highest

Table 8
Performance Analysis of Proposed Optimized model with existing work.

Reference	Model	MRI	Endoscopic	X-ray	Multi-Modality
Ayadi et al. [59]	Deep CNN	94.74	×	×	×
Mohapatra et al. [61]	Wavelet transform CNN	×	94.25	×	×
Ahmed et al. [62]	Deep CNN	×	×	90.64	×
Swati et al. [60]	VGG19	94.82	×	×	×
Nahiduzzaman et al. [63]	Lightweight CNN	×	89.13	×	×
Kör et al. [64]	CNN	×	×	97.78	×
Tehsin et al. [65]	DaSAM	85.00	×	×	×
Our	Proposed Optimized Model	94.92	95.09	97.28	97.37

scores across accuracy (97.37 %), precision (97.31 %), and recall (96.96 %), although its F1-score (96.08 %) is slightly lower than the highest F1-score achieved by MobileNetV1 (95.86 %). Among the base models, MobileNetV1 shows the best overall balance across the metrics, whereas models like MobileNetSV3 and MobileNetLV3 show relatively lower performance, particularly in precision and recall. DenseNet and Inception models present consistent performance, with DenseNet121 notably having better recall (93.75 %) compared to other DenseNet. ResNet and VGG models have lower performance metrics overall, with ResNet models struggling particularly with precision and recall, and VGG models showing some consistency but not excelling in any metrics. Overall, the proposed model demonstrates highest capability in most metrics, suggesting significant improvements over the standard pre-trained models listed.

4.8. Performance comparison of proposed model with existing models

Developing a single model for multiple medical disease diagnoses enables efficient and faster detection, reducing the need for separate

models and saving computational resources. Additionally, it enhances the scalability and robustness by providing comprehensive diagnostic capabilities in diverse medical conditions. Table 8 presents the Performance analysis of proposed optimized model with existing SOTA. Ayadi et al. [59] and Swati et al. [60] showcase high performance with Deep CNN and VGG19 models, achieving accuracies of 94.74 % and 94.82 % in MRI, respectively. However, they do not work for other modality such as endoscopic and X-ray modalities. Mohapatra et al. [61] achieved a notable accuracy of 94.25 % in endoscopic imaging using a wavelet transform CNN, but their model lacks results in MRI and X-ray applications, indicating a limitation in cross-modality performance. Ahmed et al. [62] focuses on X-ray imaging with a Deep CNN model, yielding an accuracy of 90.64 %. In contrast, Nahiduzzaman et al. [63] and Kör et al. [64] emphasize lightweight CNN and standard CNN models, with Kör et al. [64] achieving the highest accuracy of 97.78 % for X-ray imaging. Tehsin et al. [65] presented the DaSAM model with an accuracy of 85.00 %, which is lower than the other models presented, indicating potential areas for improvement in feature extraction. In contrast, our proposed optimized model outperforms the previous works with accuracy rates of 94.92 % for MRI, 95.09 % for endoscopic, and 97.28 % for X-ray, demonstrating robust multi-modality capabilities. In addition, our model can effectively work across multiple medical modalities (accuracy of 97.37 %), unlike most prior models that focus on a single dataset type. It incorporates multi-scale feature learning, enabling it to extract relevant features from various levels of the data, which is often absent in previous approaches. The combination of these capabilities contributes significantly to the accuracy and reliability of our model, making it a more robust and effective solution for medical image analysis.

4.9. Interpretability and visualization analysis

In this section, we demonstrate the effectiveness of the proposed model through interpretability and visualization analyses, including Grad-CAM [66,67] and feature map [68] analysis, compared to other models. Grad-CAM analysis is utilized to provide deeper insights into the model behavior, highlighting the features it prioritizes for

GRAD CAM Visualization Analysis Proposed Model

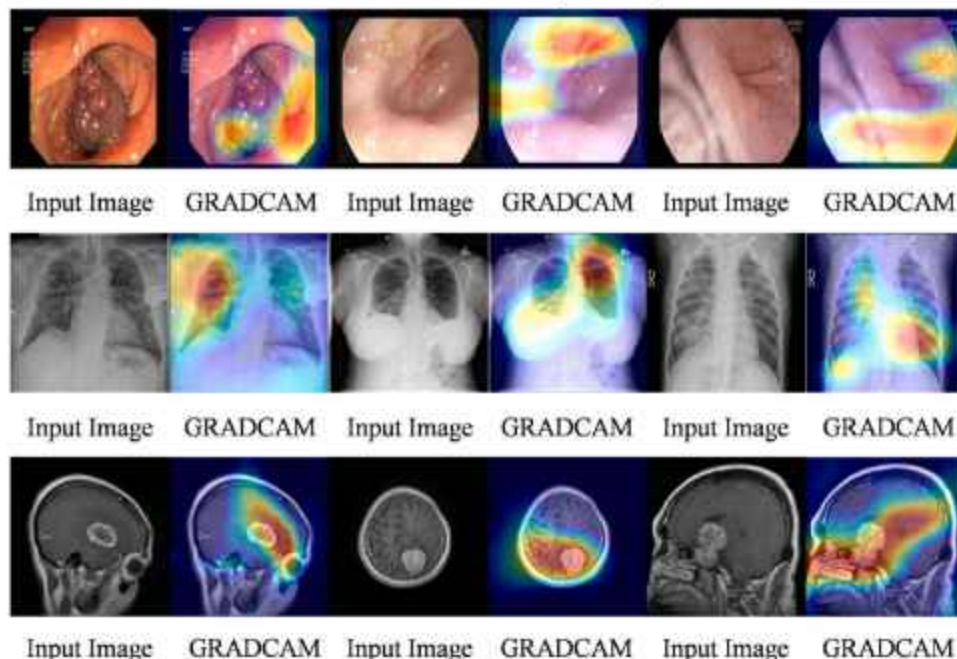


Fig. 11. GRAD CAM Visualizations Analysis.

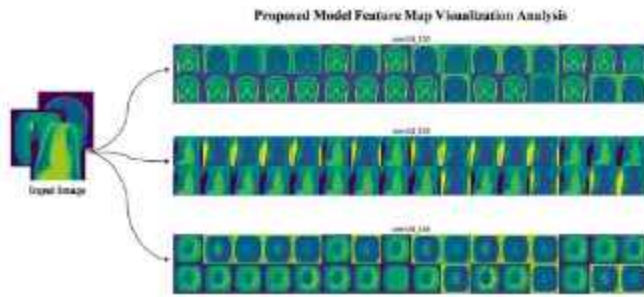


Fig. 12. Feature Map Visualizations Analysis.

decision-making. Interpretability visualization allows for visual assessment of the model interpretation of medical images. We utilized Grad-CAM analysis across multiple diverse medical datasets, including endoscopy, Chest X-ray, and MRI images. The importance of aligning DL model interpretations with expert medical insights is increasingly recognized in medical imaging research. Grad-CAM provides localized heatmaps that enhance physicians understanding of diagnostic recommendations by clearly highlighting the specific regions within medical images that influence the model decisions. In addition, Healthcare practitioners can enhance their trust in AI systems [1] by visually confirming that the model's focus aligns with relevant clinical features and ensuring that the model's reasoning is consistent with key medical insights. Grad-CAM visual feedback can enhance the training and refinement of models for clinical applications by indicating whether the model focuses on irrelevant features. This guidance helps improve both accuracy and reliability by ensuring that the model attention aligns with medically relevant aspects. The Grad-CAM (Fig. 11) results demonstrate that the proposed model effectively focuses on critical regions of the images, particularly those indicative of anomalies or infections. This selective attention confirms that the model efficiently filters out irrelevant features while accurately targeting areas characteristic of specific conditions or diseases. Such behavior underscores the success of the proposed method in accurately identifying and classifying affected areas. The model lightweight structure, multi-scale feature learning (MSFL), and optimized architecture enable it to concentrate on relevant details and omit extraneous information. Additionally, the interpretability and precise localization provided by Grad-CAM heatmaps further enhance the robustness and effectiveness of the proposed architecture. Overall, the model's ability to focus on essential information and

exclude noise significantly [69] impacts its performance across various medical imaging tasks.

Many studies [42,68] emphasize the importance of feature map analysis in understanding feature extraction processes during model training. A feature map is a vital component in neural networks [70] that captures the underlying patterns and textures from the input image. By analyzing these maps, researchers can gain insights into how a model interprets and prioritizes different features throughout the training process. This analysis not only helps in fine-tuning the model by adjusting hyperparameters and layer configurations but also enhances the transparency of model learning, making them more interpretable. Through detailed visualization of feature maps, one can observe the activation regions and identify which characteristics of the input data trigger certain behaviors in the neural network, leading to more targeted improvements in model robustness and efficacy. Fig. 12 illustrates the feature maps generated by the proposed model at various layers during training.

4.9.1. Clinical Applications

Grad-CAM and feature maps are helpful in building trust among clinicians by making the model decision-making process transparent. By visually highlighting the areas in medical images that influence the model predictions, these tools assure clinicians that the model assessments are based on relevant medical features rather than artifacts or noise. This transparency is crucial in clinical practice where trust in automated tools must be established before they are integrated into diagnostic workflows. Moreover, in complex medical cases, where diagnostic ambiguity is prevalent, interpretability tools like GradCAM can describe refined features that are often overlooked. For instance, Grad-CAM visualizations can pinpoint areas of interest in medical images that guide clinicians to focus their assessments, facilitating a more targeted approach to diagnosis. This capability is particularly valuable in cases like early-stage diseases or conditions with overlapping symptoms, where precise localization of anomalies can significantly influence treatment decisions. We further elaborate on how these GRADCAM can be systematically incorporated into the diagnostic process. For example, by integrating Grad-CAM visualizations into the review interface, clinicians can immediately observe the rationale behind a model prediction. This feature acts as a second-check mechanism, allowing healthcare providers to confirm or question the AI-generated insights with a higher degree of confidence, thus supporting more informed decision-making.

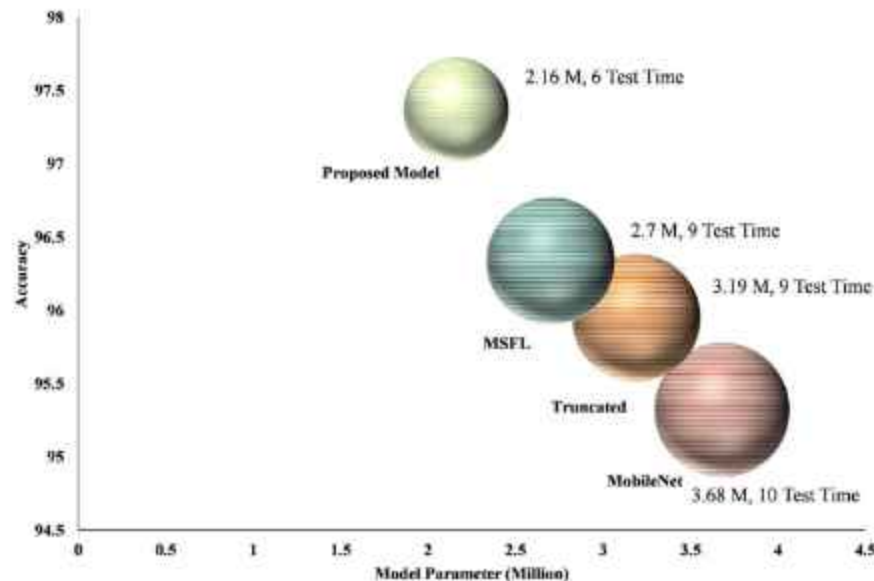


Fig. 13. Overview of Model Efficacy, and Complexity between MobileNet, Truncated, MSFL, and Proposed Model.

Table 9

Parameter Complexity Analysis of Proposed Optimized model with existing lightweight models.

Reference	Model	Parameter
Howard et al. [71]	MobileNetV3 Large	5.40 M
Howard et al. [71]	MobileNetV3 Small	2.90 M
Cocci et al. [72]	ShuffleNetV2	2.40 M
Zhou et al. [73]	GhostNet	2.56 M
Howard et al. [45]	MobileNetV1	3.68 M
Our	Proposed Optimized Model	2.16 M

For the clinical workflows, proposed model GRAD-CAM can enhance clinical decision-making by providing visual explanations of diagnostic predictions, making it easier for clinicians to understand and trust the model assessments. Additionally, the lightweight nature of our proposed model facilitates its deployment on smart devices, enabling real-time, on-site diagnostic support in clinical settings.

4.10. Complexity analysis

This section divides into the cost complexity analysis [54], an essential aspect of evaluating the efficiency of proposed model with different configuration in computational terms. It focuses on determining the trade-offs between proposed model high accuracy and resource consumption, such as time, parameters and memory. By separating the computational complexity, we can understand how changes in the input image size or parameters affect the performance and scalability of proposed model. This analysis is crucial for optimizing models, especially in environments with limited resources, ensuring proposed model deliver desired outputs without excessive computational costs.

Fig. 13 presents the accuracy, test time for classifying images into their respective classes, model parameters (M) required for storage and memory load between the proposed model and its different configurations. The proposed model achieved the highest accuracy with a modest number of parameters, 2.16 million, and a test time of 6 seconds. It reduces the parameters by 1.52 million compared to its original MobileNet base model. This reduction not only requires fewer parameters but also achieves higher accuracy. Additionally, the proposed model efficiently integrates the modified naive inception block in its multi-scale feature module to address overfitting issues. The proposed model distinguishes from the baseline by striking an optimal balance between accuracy and complexity. It presents a practical and efficient solution for medical image classification tasks, delivering high accuracy while maintaining a manageable parameter count. This model underscores the importance of considering factors beyond accuracy, such as resource utilization and practical applicability, when evaluating and comparing models in medical contexts.

To address the concern regarding model complexity and overfitting, the proposed model emphasizes several strategies that mitigate overfitting risks despite the complexity of the architecture. The model

Table 10

Class wise Performance Analysis of Proposed Optimized mode on Additional test dataset.

Models	Class	Precision	Recall	F1-score	Accuracy
Proposed Optimized Model	Colon Aca	0.99	1.00	1.00	97.33
	Colon N	1.00	0.99	0.99	
	Colorectal Cancer	1.00	1.00	1.00	
	COVID	0.95	0.97	0.96	
	Esophagitis	0.82	0.86	0.84	
	Glioma	0.94	0.92	0.93	
	Lung Opacity	0.97	0.95	0.96	
	Meningioma	0.91	0.90	0.90	
	Pituitary	0.96	0.99	0.97	
	Pylorus	0.89	0.90	0.84	
Proposed Optimized Model	Viral	0.99	0.99	0.99	
	Pneumonia Benign	0.91	0.89	0.85	97.31
	Colon Aca	0.99	0.99	0.99	
	Colon N	0.99	0.99	0.99	
	Colorectal Cancer	0.93	0.96	0.95	
	COVID	0.93	0.97	0.95	
	Esophagitis	0.88	0.77	0.86	
	Glioma	0.97	0.93	0.95	
	Lung Opacity	0.97	0.93	0.95	
	Malignant	0.82	0.77	0.73	
	Meningioma	0.87	0.91	0.89	
	Pituitary	0.95	0.99	0.97	
	Pylorus	0.89	0.90	0.87	
	Viral	0.99	0.99	0.99	
	Pneumonia				

incorporates a modified naive inception block within its multi-scale feature module, which is designed specifically to enhance generalization and avoid overfitting by capturing diverse image features at different scales. This approach helps the model focus on critical features while reducing redundancy in feature extraction. Additionally, the use of Mid truncation in the MobileNet architecture reduces the overall model complexity by strategically removing fewer effective layers, which not only decreases the number of parameters but also minimizes the risk of overfitting by preventing the model from memorizing data patterns that are too specific. Furthermore, the integration of meta-heuristic optimization techniques, assists in refining the truncated model architecture and parameter selection through iterative optimization rounds. This process helps the model avoid overfitting by continually adjusting the parameters to prevent over-specialization on the training data. Finally, the relatively less parameter count of the proposed model, 2.16 million, combined with an efficient optimization process, ensures that the model maintains a balance between high performance and generalization, particularly in the context of relatively small medical image datasets.

Table 9 presents a parameter complexity analysis comparing our

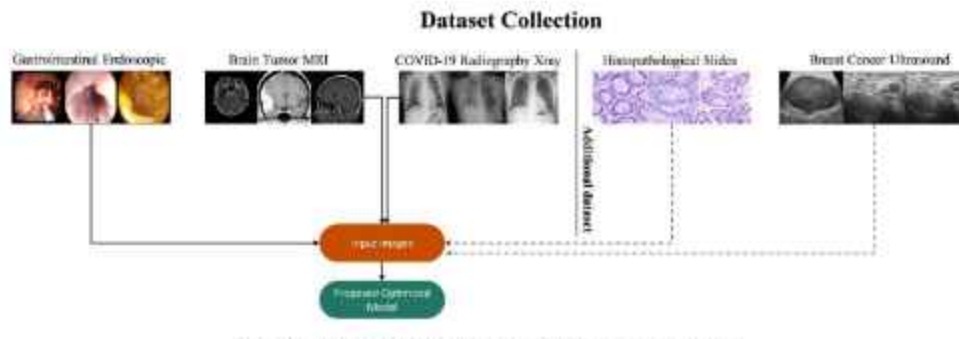


Fig. 14. Dataset Samples Images of each dataset modality.

Proposed Optimized Model with MRFO Metaheuristic Confusion Matrix Analysis

Actual Values	Colon ACA	Colon N	Colorectal cancer	COVID	Esophagitis	Glioma	Lung Opacity	Meningioma	Pituitary	Pylorus	Viral Pneumonia
	399	0	0	0	0	0	0	0	0	0	1
	1	399	0	0	9	0	0	0	0	0	0
	0	0	28	0	0	0	0	0	0	0	0
	0	0	0	396	0	0	4	0	0	0	0
	0	0	0	0	17	0	0	0	0	4	0
	0	0	0	0	0	285	0	13	2	0	0
	0	0	0	11	0	0	387	1	0	0	1
	0	0	0	1	0	16	0	277	11	0	1
	0	0	3	0	0	0	0	3	294	0	0
	0	0	0	0	3	0	0	0	0	27	0
	0	0	0	0	0	0	0	0	0	0	269
Predicted Values											

Fig. 15. Confusion Matrix Visualization with additional test.

proposed optimized model with existing lightweight models. The parameter count is a critical factor in determining the efficiency and performance of neural networks, especially in mobile and edge computing applications. Howard et al. [71] introduced both MobileNetV3 Large and MobileNetV3 Small, which have parameter counts of 5.40 million and 2.90 million, respectively. These models are designed for various applications, but their parameter counts remain relatively high. Cococi et al. [72] introduces ShuffleNetV2, which has a parameter count of 2.40 million, while Zhou et al. [73] presents GhostNet with 2.56 million parameters. Both models are known for their efficiency but still have a higher parameter count than our proposed model. Howard et al. [45] provides data on MobileNetV1, which has 3.68 million parameters, further emphasizing the trend of lightweight models aiming for lower complexity. In contrast, our proposed optimized model achieves a remarkable parameter count of 2.16 million, making it the most efficient model in this comparison. Due to optimal model truncation and compression such as Mid truncation, our proposed optimized model significantly reduces the parameter count compared to other lightweight models. This reduction enhances efficiency while maintaining robust performance, making it an optimal choice for deployment in resource-constrained environments.

4.11. Additional test

In this research, we have carried out more tests to further illustrate the effectiveness and uniqueness of the proposed one model for multiple

medical disease diagnosis. Verifying model performance with more test datasets improves the result generalizability and makes sure the model can handle a range of dataset modality. By detecting any possible overfitting to the training set, this procedure enhance clarification in the model performance in real-world situations. The Histogram Pathological dataset [74], and Breast Ultrasound Images [75] are two more dataset that we have used in this research as an additional dataset. Sample images from utilized datasets are shown in Fig. 14. The experimental setup for our first multi-dataset modality disease detection framework employed consistent configurations for model training (80 %), validation (10 %), and testing (20 %). The optimized model utilized the Adam optimizer, trained over 30 epochs, with a batch size of 16 and a learning rate set to 0.001. These uniform parameters enabled a direct comparison of performance across additional dataset training and evaluation phases.

The proposed optimized model shows minimal accuracy loss with the addition of the new dataset, dropping slightly from 97.38 % (with 9 diseases) to 97.33 % (with 11 diseases). This demonstrates its robustness and ability to generalize well across diverse disease categories. The consistent high performance across various classes illustrates the model effectiveness in multi-disease detection, making it a valuable tool for clinical applications. Table 10 presents that model demonstrates high precision, recall, and F1-scores across most classes, notably achieving accurate scores (1.00) in the Colon Aca, Colorectal Cancer, Colon N, and Viral Pneumonia categories. Beside of this, there are some variances in performance, particularly in the Esophagitis category, where the precision is notably lower at 0.82 and results in an F1-score of 0.84. This

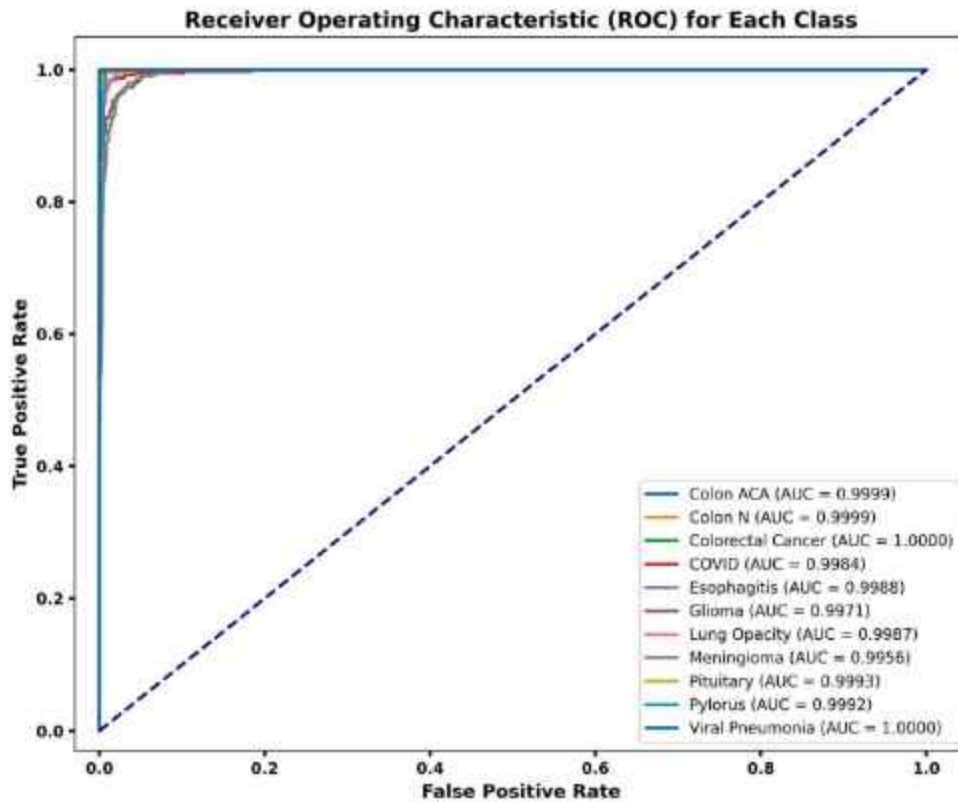


Fig. 16. Class wise Performance Analysis (AUC-ROC) of Proposed Model on additional dataset.

indicates a challenge in correctly identifying cases of esophagitis compared to other conditions. In addition, model maintains high performance in Glioma (F1-score of 0.93) and Lung Opacity (F1-score of 0.96), showcasing its robustness in identifying these diseases. Moreover, Table 10 presents that slightly reduce in the results further when ultrasound breast cancer dataset modality integrates in it. The model achieves excellent performance for conditions like Colon ACA, Colon N, and Viral Pneumonia, with highest scores across all three metrics (0.99). For conditions like Colorectal Cancer, COVID, Glioma, and Lung Opacity, the model also shows notable performance with scores mostly above 0.93. However, there are notable discrepancies in conditions such as Esophagitis, Malignant, and Meningioma, where the metrics drop below 0.90, indicating less reliability in those classifications. Particularly, the Malignant category shows significantly lower scores in all metrics (below 0.82), suggesting the model struggles the most with this condition. In contrast, Pituitary tumors are well-identified, evidenced by high precision (0.99) and efficient overall scores, pointing to the model varying effectiveness depending on the complexity and characteristics of the conditions being diagnosed.

In Fig. 15, the confusion matrix illustrates the performance of the proposed optimized model with MRFO for eleven disease classification, showing high accuracy across various disease classes. The model has accurately classified Colon ACA 399 test images out of 400, COVID 396 test images out of 400, and Viral Pneumonia 269 test images achieving notable accuracy. Misclassifications, such as Colon N incorrectly classifying 1 image as Colon ACA and 9 as Colorectal Cancer, and Esophagitis misclassifying 17 test images as Glioma, indicate areas for potential improvement. Overall, the proposed optimized model demonstrates an effective capability to accurately differentiate between multiple diseases, highlighting its effectiveness for clinical applications in disease detection.

Fig. 16 presents the AUC-ROC curve analysis of each class in dataset. Visualizing the AUC-ROC curve for each class in the proposed model offers a strong visual representation of the model ability to distinguish

between classes, helping to assess its performance and identify specific areas where improvements might be needed. The results demonstrate that the optimized model excels in accurately classifying test images for specific conditions, achieving scores (AUC of 1.00) for both Colorectal Cancer and Viral Pneumonia. This indicates that the model can reliably distinguish between positive and negative cases in these categories without any errors. On the other hand, for Colon ACA and Colon N, the model exhibited minor discrepancies, misclassifying one test image in each category, which resulted in a slightly reduced AUC of 0.99 for these classes. This suggests a high level of accuracy, though with slight room for improvement in distinguishing these conditions. The most challenging category for the model was meningioma brain tumors, where it achieved an AUC of 99.56, reflecting the model struggle with the inherent complexity and variability of these tumor images, indicating that further refinement might be needed to enhance its diagnostic precision in this area.

Fig. 17 presents a confusion matrix that evaluates the performance of the proposed optimized model with MRFO in classifying thirteen diseases. The model shows high accuracy across different disease categories, successfully identifying all 28 test images of Colorectal Cancer, and correctly classifying 397 out of 400 test images for both Colon ACA and Colon N. Notable misclassifications include Glioma, where 24 images were incorrectly classified as Meningioma and Pituitary, and Esophagitis, with 3 images misclassified as Glioma and Malignant. These results point to areas where the model could be further improved. Overall, the model effectively distinguishes between multiple diseases, underscoring its potential utility in clinical settings for disease detection.

Fig. 18 displays the AUC-ROC curve analysis for each class in the dataset. The optimized model shows exceptional performance in accurately classifying test images for conditions such as Colon ACA, Colorectal Cancer, and Colon N, achieving perfect AUC scores of 1.00. This performance underscores the model's capability to flawlessly differentiate between positive and negative cases in these categories. Conversely, the model encounters some challenges with Glioma,

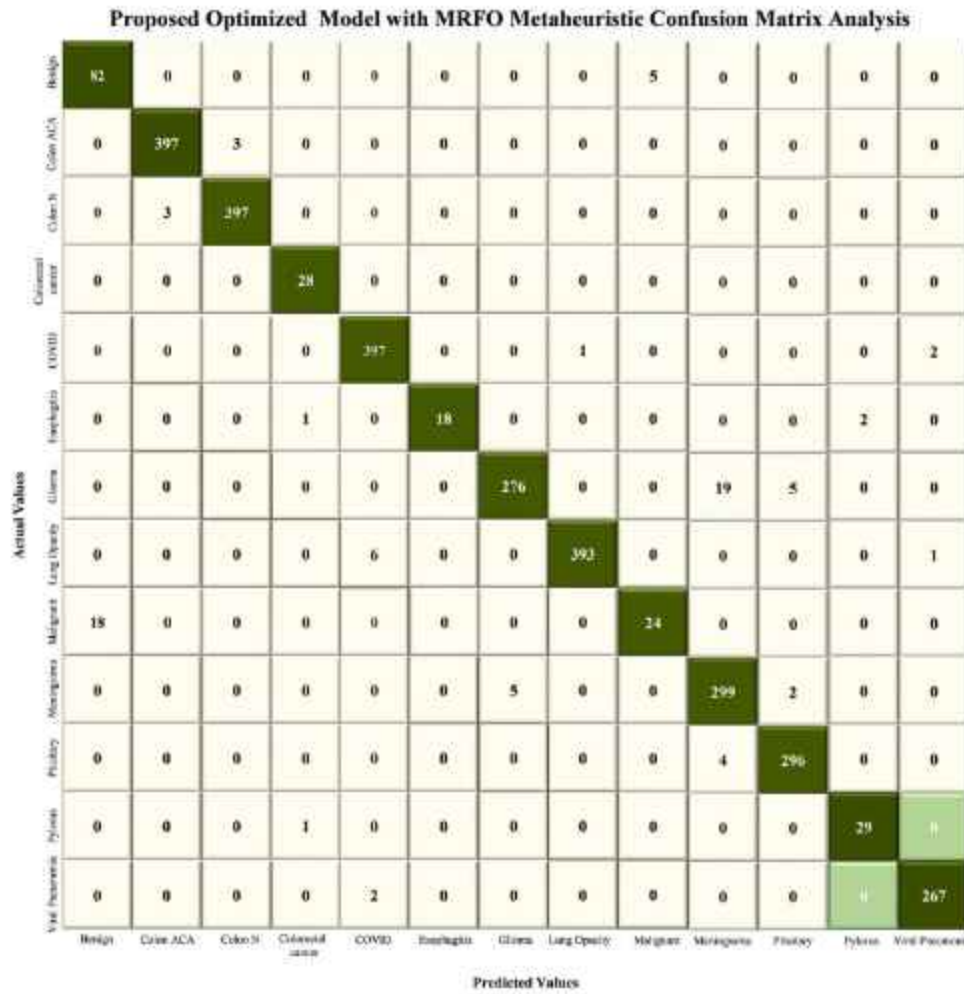


Fig. 17. Confusion Matrix Visualization with additional test.

Malignant, and Meningioma categories, misclassifying one test image in each, resulting in slightly reduced AUC scores of 99.65, 99.51, and 99.49, respectively. These results indicate a high accuracy level but suggest that further improvements could enhance the model's discriminatory ability in these more complex conditions. Meningioma brain tumors, in particular, present the greatest challenge, with an AUC of 99.49, reflecting the model's struggle with the complexity and variability inherent in these tumor images and highlighting the need for additional refinement to improve its diagnostic accuracy.

4.12. Ablation study

In this section, we present the detail discussion about proposed model, and performance effect of its different architecture configuration. Previous studies shows that ablation studies on proposed model and its different configurations offer several benefits. Firstly, study provide insight into the specific contributions of each component, highlighting how various configurations impact the overall performance and guiding further optimizations. Secondly, these studies serve as a validation tool for the model design by systematically removing or modifying components to confirm the necessity and efficiency of each architectural element. Lastly, studies enhance the model robustness by identifying weaknesses or over-dependencies on certain features, enabling the development of models that maintain consistent performance across diverse conditions and datasets.

Table 11 presents a comprehensive comparison of the performance

of an Inception Block model under various modifications and settings, measured across several key metrics: Accuracy, Precision, Recall, and F1-score. Each section of the table tests different variables, such as learning rates, batch sizes, epochs, and optimizers, to ascertain their effect on model performance. Initially, adjusting the learning rate (LR) shows a notable variance in outcomes. The model achieves high performance with an LR of 0.001, achieving 97.37 % accuracy and corresponding high precision, recall, and F1-scores of over 96 %. At contrary, a high LR of 0.1 significantly drops performance, whereas a moderate LR of 0.01 maintains consistent results. Batch size variations (BS) indicate that smaller sizes (8 and 16) lead to slightly better results than larger ones (32 and 64), maintaining above 90 % in all metrics, suggesting that smaller batches optimize learning. Different epochs (EP) show that increasing beyond 20 doesn't necessarily improve results; at 20 epochs and then slightly drop with further increases. Lastly, when testing optimizers, ADAM stands out with the highest scores across all metrics, confirming its efficacy, while others like Adadelta and Adagrad show significantly weaker performance, particularly impacting accuracy and F1-scores. This comprehensive testing helps in understanding optimal model configurations for achieving the best performance.

The performance analysis presented in the Table 12 highlights the improvements achieved by each model iteration. The MobileNetV1-Base model achieved an accuracy of 95.32, with precision, recall, and F1-score values of 94.22, 93.82, and 94.05, respectively. The mid-truncated model demonstrated an increase in accuracy to 95.95, with slight enhancements in precision and recall, achieving 94.81 and 94.95.

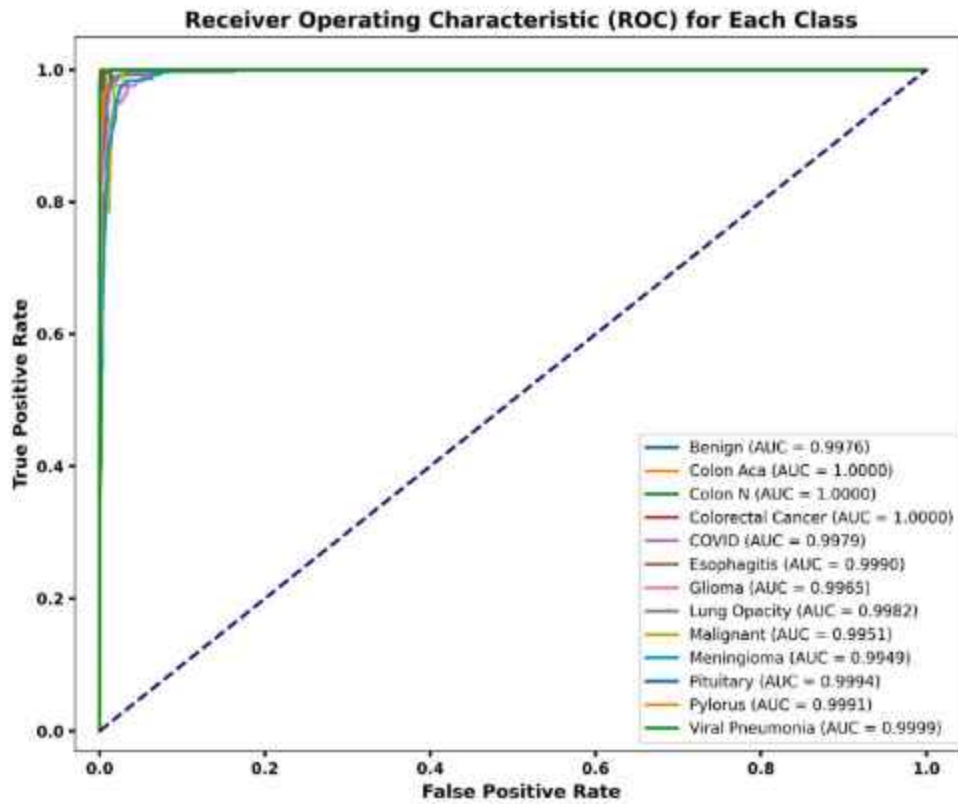


Fig. 18. Class wise Performance Analysis (AUC-ROC) of Proposed Model on additional dataset.

Table 11

Blocks wise testing of MobileNetV1 (LR) Architecture.

Models	Accuracy	Precision	Recall	F1-score
Inception Block	95.22	91.42	90.82	91.05
Inception Block with LR	97.37	97.31	96.96	96.08
Inception Block with LR (DW)	95.56	94.21	92.76	93.09
With Different Learning Rates (EP=20, BS=16)				
Inception Block with LR (0.1)	76.92	68.26	64.02	58.55
Inception Block with LR (0.01)	93.57	91.41	91.24	91.28
Inception Block with LR (0.001)	97.37	97.31	96.96	96.08
Inception Block with LR (0.0001)	91.57	89.64	89.59	89.59
With Different Batch Sizes (LR=0.001, EP=20)				
Inception Block with BS (8)	94.64	92.02	91.78	91.70
Inception Block with BS (16)	96.05	93.31	92.96	93.08
Inception Block with BS (32)	95.37	93.43	92.87	93.03
Inception Block with BS (64)	93.42	90.95	90.67	90.75
With Different Epochs (LR=0.001, BS=16)				
Inception Block with EP (20)	97.37	97.31	96.96	96.08
Inception Block with EP (30)	95.37	93.95	93.94	93.89
Inception Block with EP (40)	95.81	93.90	92.98	93.21
Inception Block with EP (50)	95.95	94.16	93.83	93.93
With Different Optimizers (LR=0.001, BS=16, EP=200)				
Inception Block with ADAM	97.37	97.31	96.96	96.08
Inception Block with RMSprop	94.83	92.85	92.93	92.75
SGD	95.18	93.20	92.79	92.93
Adadelta	78.96	80.85	78.22	78.41
Adagrad	86.07	86.55	86.32	86.38
Adamax	92.98	90.75	90.54	90.51

The MSFL model further improved the accuracy to 96.34, along with precision and recall values of 96.11 and 95.69, indicating effective multi-scale learning.

Integrating ConvLSTM into the MSFL model resulted in a notable accuracy increase to 96.76, with precision at 95.33 and recall at 94.89. Finally, the proposed optimized model reached the highest performance, achieving an accuracy of 97.37, with precision, recall, and F1-score

Table 12

Performance Analysis of MobileNetV1-Base, Mid Truncated Model, MSFL, MSFL Model with ConvLSTM, and the Proposed Optimized Model.

Model	Accuracy	Precision	Recall	F1-score
MobileNetV1-Base	95.32	94.22	93.82	94.05
Truncated Model (Mid)	95.95	94.81	94.95	94.52
MSFL Model	96.34	96.11	95.69	95.86
MSFL with ConvLSTM	96.76	95.33	94.89	95.40
Proposed Optimized Model	97.37	96.73	95.42	96.53

Table 13

Breakdown of the Proposed Optimized model Performance on each individual dataset.

Models	Class	Precision	Recall	F1-score	Accuracy
Proposed Optimized Model	Glioma	0.98	0.91	0.95	94.92
	Meningioma	0.93	0.94	0.93	
	Pituitary	0.94	0.99	0.97	
Proposed Optimized Model	Colorectal Cancer	1.00	0.89	0.94	95.09
	Esophagitis	0.94	0.93	0.93	
	Pylorus	0.92	0.95	0.93	
Proposed Optimized Model	COVID	0.95	0.98	0.97	97.28
	Lung Opacity	0.98	0.95	0.97	
	Viral Pneumonia	0.99	1.00	0.99	
	Pneumonia				

values of 96.73, 95.42, and 96.53, respectively. Evaluating each improvement from the base model to the final optimized model is essential to understand the effectiveness of each enhancement and to ensure that the modifications contribute positively to the model overall performance. This systematic assessment also aids in identifying the key

Table 14

Breakdown of the Proposed model Optimized with different optimization techniques Performance on Multi Medical dataset Modality.

Models	Precision	Recall	F1-score	Accuracy
Proposed Model Optimized with PSO Algorithm	94.82	94.23	95.02	94.93
Proposed Model Optimized with Salp Swarm Algorithm	96.52	96.29	96.36	96.39
Proposed Model Optimized with Manta Ray Foraging Algorithm	96.73	95.42	96.53	97.37

factors improvements, guiding future research and development efforts.

Table 13 provides a detailed breakdown of the performance of the proposed optimized model on individual datasets related to various medical conditions. For the glioma class, the model achieved a precision of 0.98, recall of 0.91, F1-score of 0.95, and an accuracy of 94.92. In the meningioma class, the model demonstrated a precision of 0.93 and recall of 0.94, resulting in an F1-score of 0.93. The pituitary class performed similarly well, with a precision of 0.94, a high recall of 0.99, and an F1-score of 0.97. For the colorectal cancer dataset, the model achieved perfect precision at 1.00, with a recall of 0.89 and an F1-score of 0.94, resulting in an accuracy of 95.09. In the esophagitis and pylorus classes, the model maintained high precision and recall scores, indicating consistent performance. For the COVID dataset, the proposed model demonstrated robust performance with a precision of 0.95, recall of 0.98, F1-score of 0.97, and an accuracy of 97.28. Additionally, the model performed excellently on lung opacity and viral pneumonia classes, achieving very high precision, recall, and F1-scores. Validating the model performance on each individual dataset is crucial to ensure that it effectively addresses the unique characteristics and challenges associated with different medical conditions. This assessment proved proposed strengths and weaknesses of the model, allowing for targeted improvements and enhancing its reliability future directions.

Table 14 illustrates the performance metrics of the proposed model optimized with different optimization techniques PSO (Particle Swarm Optimization), Salp Swarm Algorithm, and Manta Ray Foraging Algorithm on Multi Medical dataset Modality. Our selected Manta Ray Foraging Algorithm outperforms the others with the highest accuracy at 97.37 % and high precision and recall, indicating a more effective optimization in refining the model predictive capabilities. The Salp Swarm Algorithm also shows competitive results, particularly in precision and recall, underscoring its efficiency in balancing model sensitivity and specificity. In contrast, the PSO Algorithm, while still delivering acceptable outcomes, shows slightly lower metrics, suggesting it may be less effective at optimizing the model across datasets.

5. Research limitation

Despite the multiple advantages of the proposed methodology, the suggested model does have some limitations. While the model performs well across the tested modalities (brain MRIs, chest x-rays, gastro endoscopic images), Its adaptability to entirely new or private medical imaging modalities has not been fully explored. Due to the acquiring sensitive nature of medical datasets, the proposed model was not evaluated in this aspect. In addition, limitation is the dependency on large, annotated datasets for training the model. While the model generalizes well to multiple disease categories, its performance may affect if applied to datasets with insufficient or noisy labels.

Future research should focus on addressing these limitations by exploring unsupervised or semi-supervised learning techniques and expanding the model's adaptability across a wider range of medical imaging tasks.

6. Conclusion and future direction

This study introduced an efficient framework, the One Model for Multiple Diseases, leveraging a lightweight MobileNet architecture and incorporating novel approaches such as model truncation and a modified naive inception block. By integrating nature inspire metaheuristic optimization techniques, specifically chain foraging and cyclone aging, the model demonstrated significant improvements in generalization across diverse image features. The integration of an ConvLSTM layer before the Softmax classification layer further enhanced the model ability to capture both temporal and spatial dimensions of features, making it highly effective for multi-disease diagnosis. Through extensive experiments on publicly available medical image datasets, the proposed framework achieved highest accuracy of 97.37 %, demonstrating its robustness, scalability, and ability to outperform existing methods.

Our research contributes to the field in several key ways. We have introduced a mid-level truncation approach, specifically the MobileNetV1 DS (-2) configuration, which consistently outperforms start and end truncation methods by preserving essential features while reducing computational overhead. In addition, the integration of metaheuristic optimization techniques, such as chain foraging and cyclone aging, represents a novel application in medical imaging, enhancing the model generalization and robustness across varying image modalities. Additionally, the use of an ConvLSTM in combination with convolutional layers presents a significant advance in modeling temporal-spatial dependencies, a crucial aspect in diagnosing multiple diseases from diverse datasets. These theoretical advancements set a new benchmark for efficient DL models in the healthcare domain. From a practical application, the proposed framework offers several significant advantages for clinical settings. The lightweight nature of the MobileNet architecture, combined with model truncation, ensures that the model operates efficiently even in resource-constrained environments, such as mobile healthcare devices or edge computing platforms. Moreover, the integration of Grad-CAM and feature map visualization techniques enhances interpretability, allowing clinicians to better understand and trust the model predictions, thus improving clinical workflow and decision-making. This balance between computational efficiency and high accuracy positions the model as a practical solution for deploying AI-powered diagnostic tools in medical settings.

For future research directions, as our study establishes a balance between maintaining low computational demands and achieving high performance relative to competing models. Moving forward, we plan to expand our research in several ways: To enhance the generalizability and effectiveness of our model, we intend to extend its application to further diverse dataset modalities. This includes experimenting with a variety of image resolutions, modalities like Histogram, or ultrasound, and even integrating non-image data such as patient demographics or clinical outcomes. We will explore cutting-edge DL methodologies that have shown promise in other fields, such as transformer models or capsule networks. By adapting these advanced techniques to our framework, we aim to improve the model accuracy, interpretability, and learning efficiency.

CRedit authorship contribution statement

Saif Ur Rehman Khan: Writing – review & editing, Writing – original draft, Visualization, Validation, Formal analysis, Data curation, Conceptualization. **Ming Zhao:** Supervision. **Sohaib Asif:** Writing – review & editing. **Wei Zou:** Writing – review & editing. **Yangfan Li:** Supervision, Writing – original draft, Writing – review & editing. **Xiangmin Li:** Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence

the work reported in this paper.

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Data availability

Data will be made available on request.

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